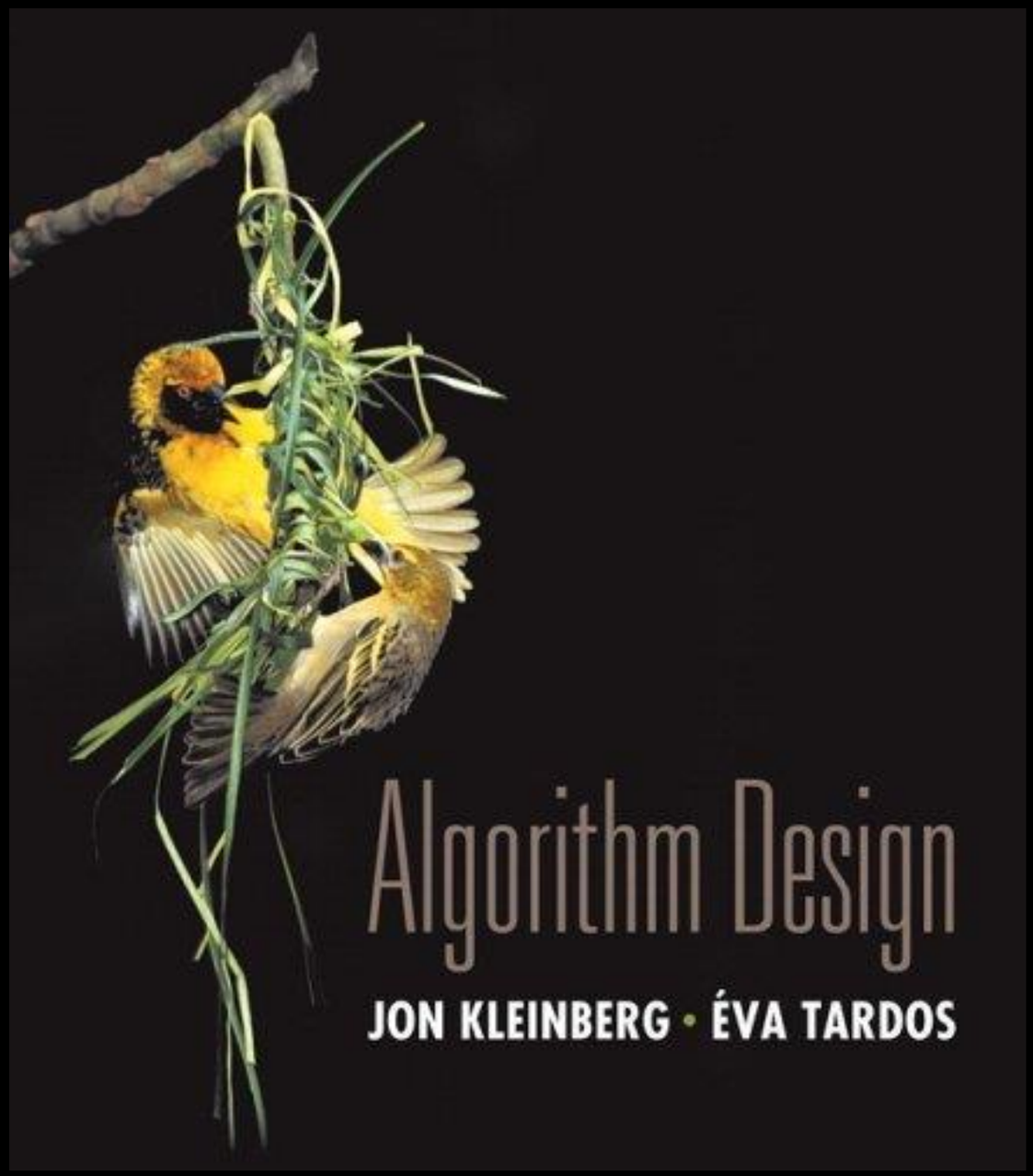

Chapter 4

Interval scheduling

Slides by Kevin Wayne.
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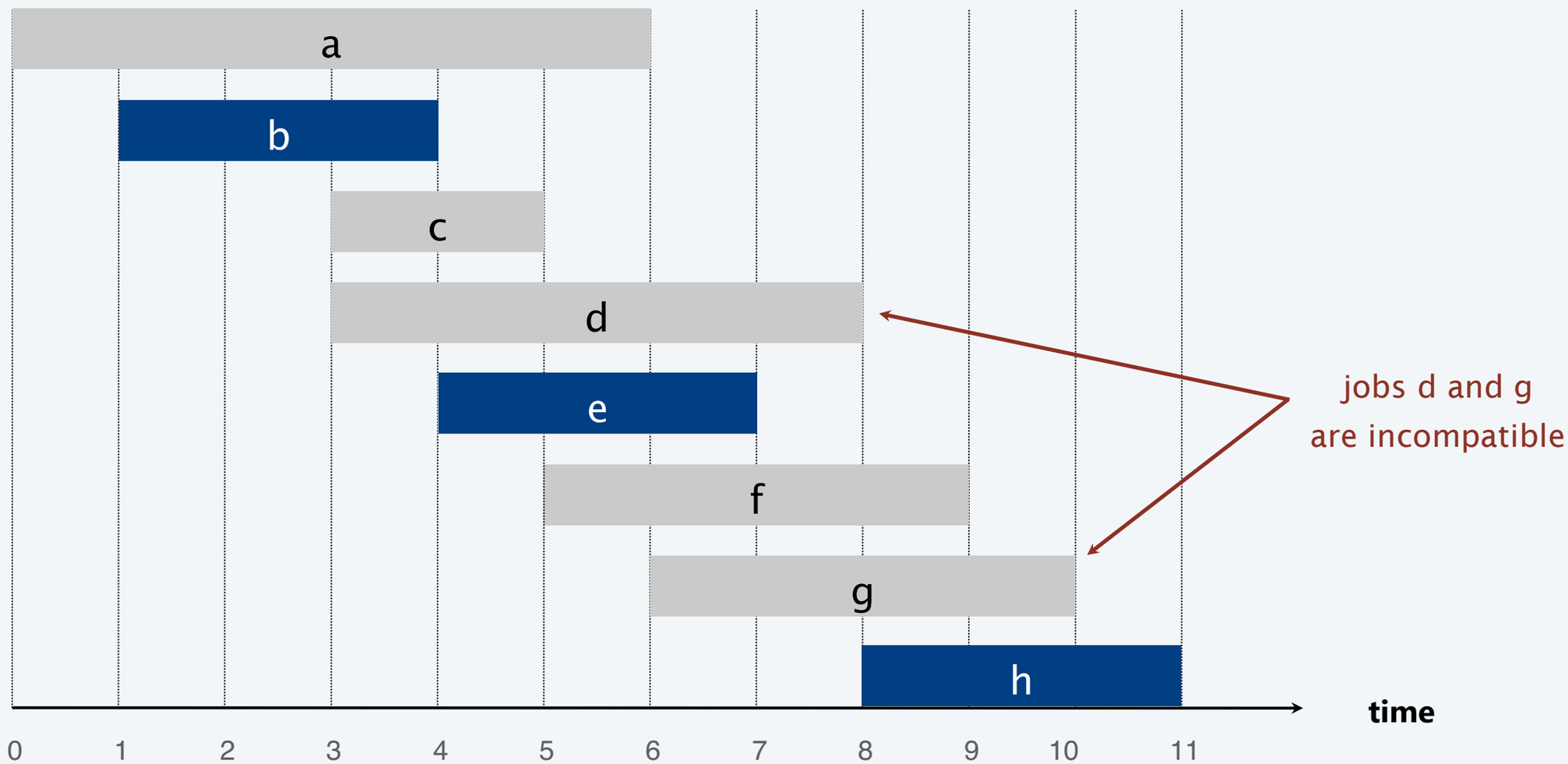
Interval scheduling

- Job j starts at s_j and finishes at f_j.
- Two jobs are compatible if they don't overlap.
- Goal: find maximum subset of mutually compatible jobs.

(cardinalità massima)



Interval scheduling

- Input:
  - A set of n intervals $I_1,\ldots,I_n$
  - interval I_i has starting time s_i and finish time f_i
- Feasible solution:
  - A subset S of the intervals that are mutually compatible, i.e. for each $I_i,I_j \in S$, I_i does not overlap with I_j
- Measure (to maximize):
  - number of scheduled intervals, i.e. cardinality of S

Interval scheduling: greedy algorithms

Greedy template. Consider jobs in some natural order.

Take each job provided it's compatible with the ones already taken.

Posso sceglierne una, ma avrò risultati differenti

- [Earliest start time] Consider jobs in ascending order of sj.

- [Earliest finish time] Consider jobs in ascending order of fj.

  il migliore

- [Shortest interval] Consider jobs in ascending order of fj – sj.

- [Fewest conflicts] For each job j, count the number of conflicting jobs cj. Schedule in ascending order of cj.

Interval scheduling: greedy algorithms

Greedy template. Consider jobs in some natural order.

Take each job provided it's compatible with the ones already taken.

counterexample for earliest start time

Ne prendo 1 ma potevo prenderne 4

counterexample for shortest interval

Ne prendo 1 ma potevo prenderne 2

counterexample for fewest conflicts

Ne prendo 3 ma potevo prenderne 4

Interval scheduling: earliest-finish-time-first algorithm

EARLIEST-FINISH-TIME-FIRST $(n, s_1, s_2, \ldots, s_n, f_1, f_2, \ldots, f_n)$

SORT jobs by finish times and renumber so that $f_1 \leq f_2 \leq \ldots \leq f_n$.

$S \leftarrow \varnothing$. ← set of jobs selected inizialmente l'insieme è vuoto

FOR $j = 1$ TO n

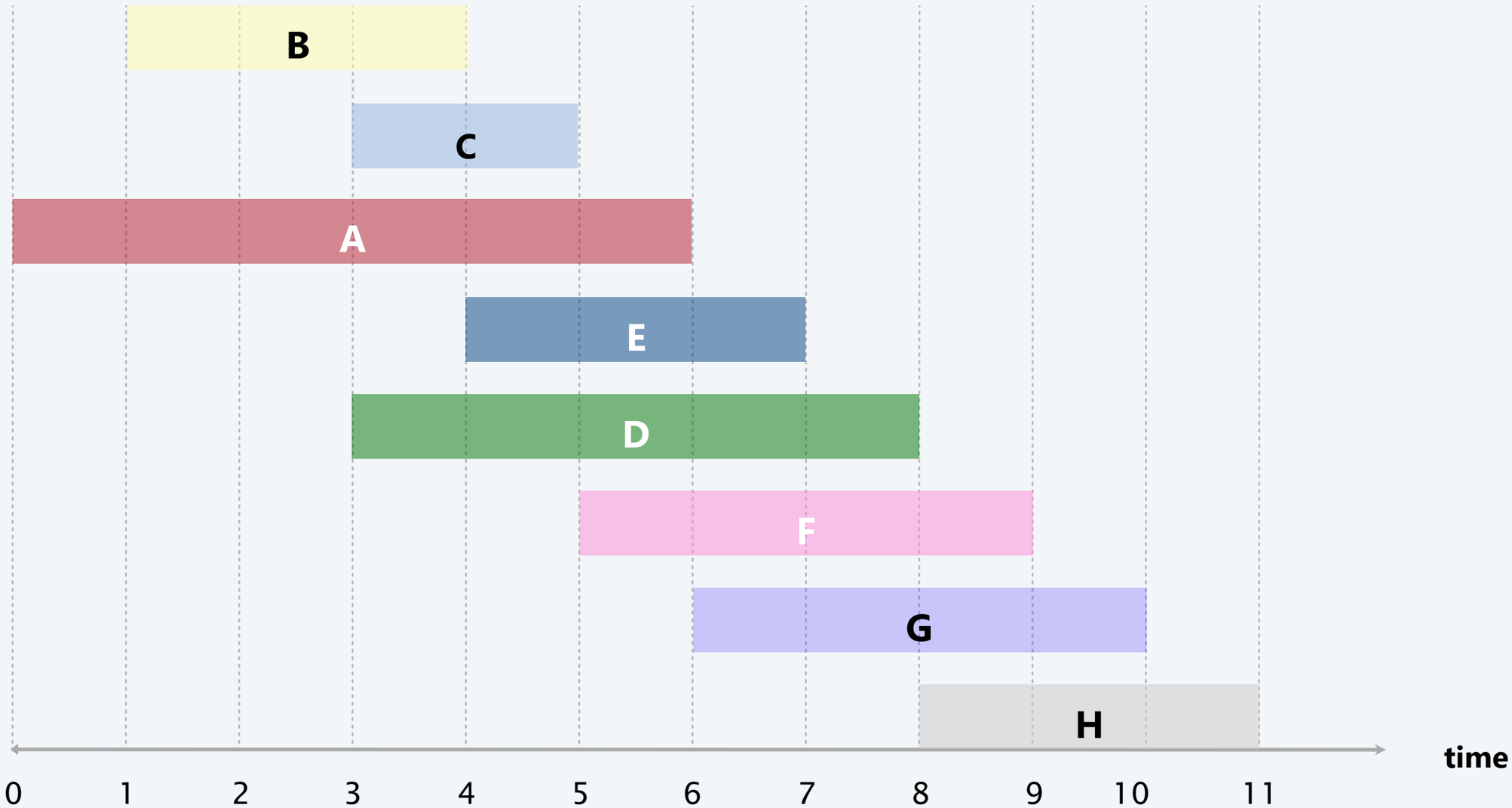
  IF (job j is compatible with S)

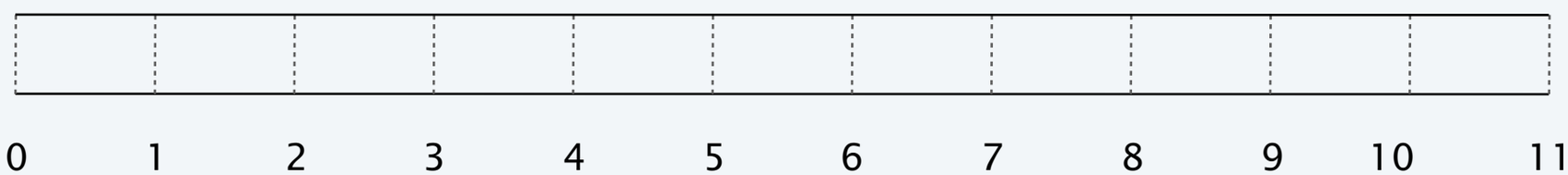
    $S \leftarrow S \cup \{ j \}$.

RETURN S.

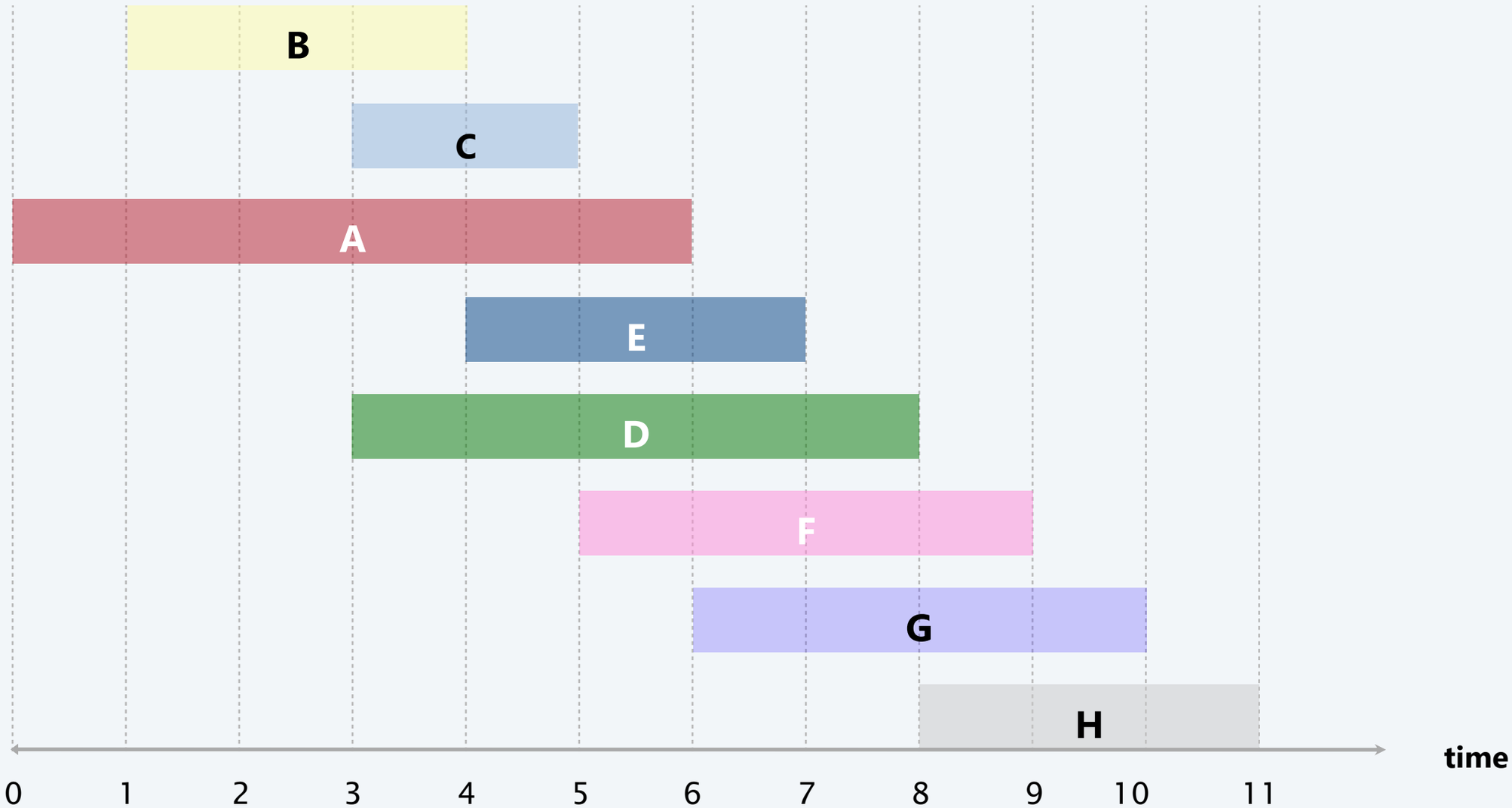
Earliest-finish-time-first algorithm demo

Per prima cosa, riordino i lavori per TEMPI DI FINE

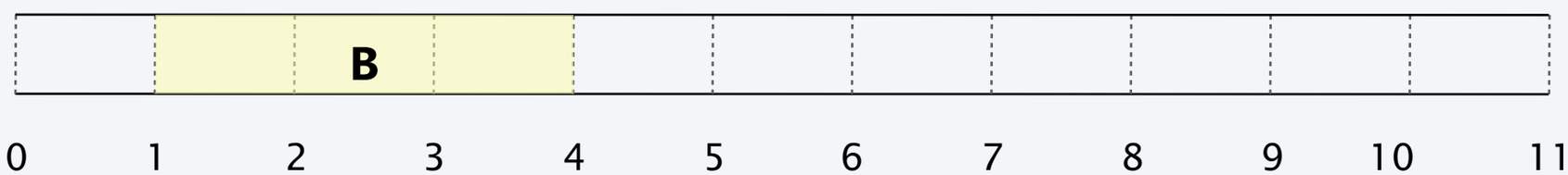




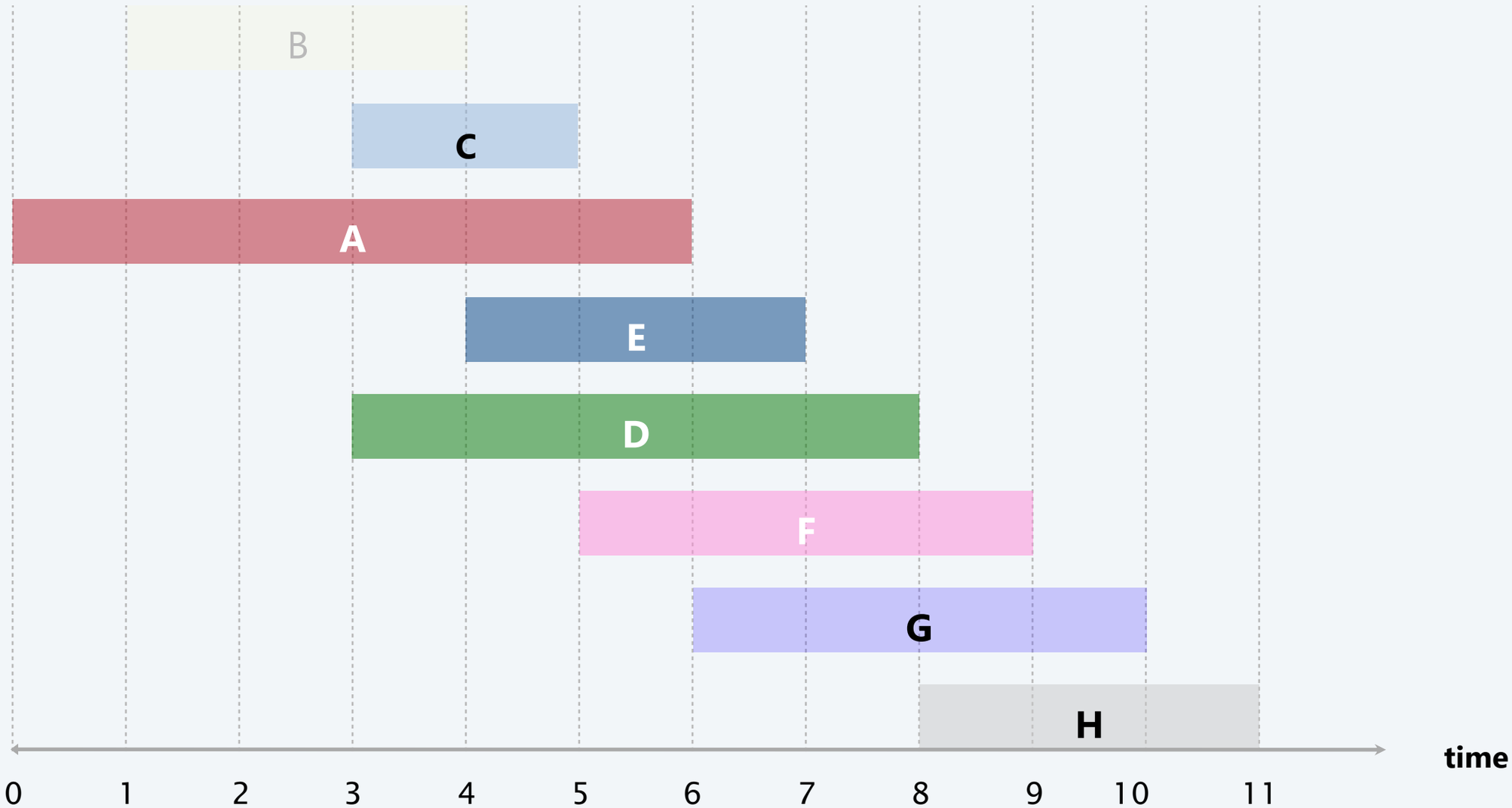
Earliest-finish-time-first algorithm demo



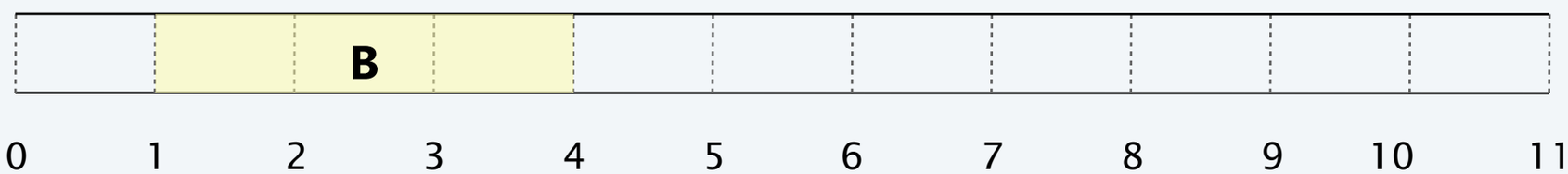
job B is compatible (add to schedule)



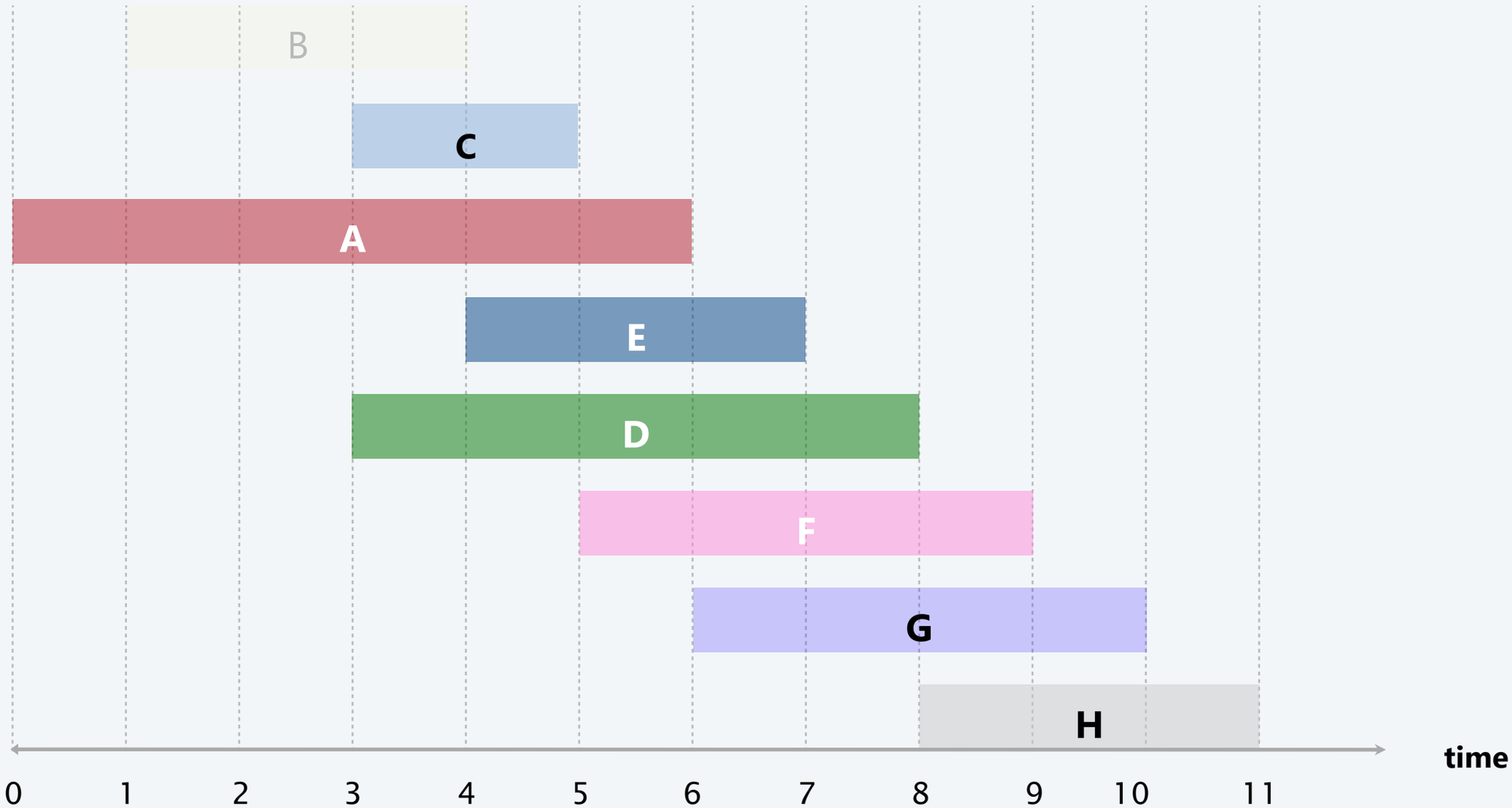
Earliest-finish-time-first algorithm demo



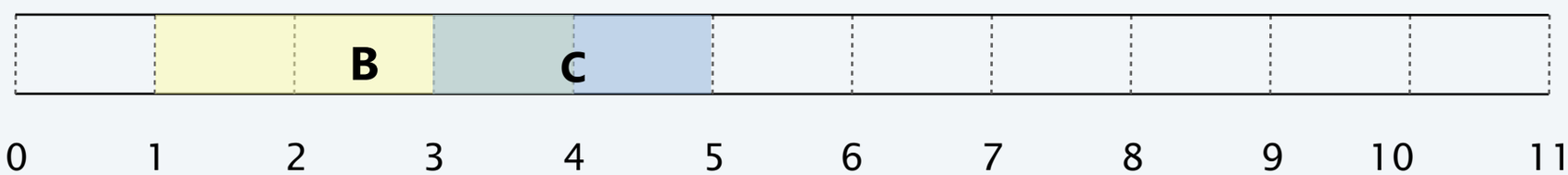
N.B. = dopo il riordinamento, il primo job è SEMPRE da prendere



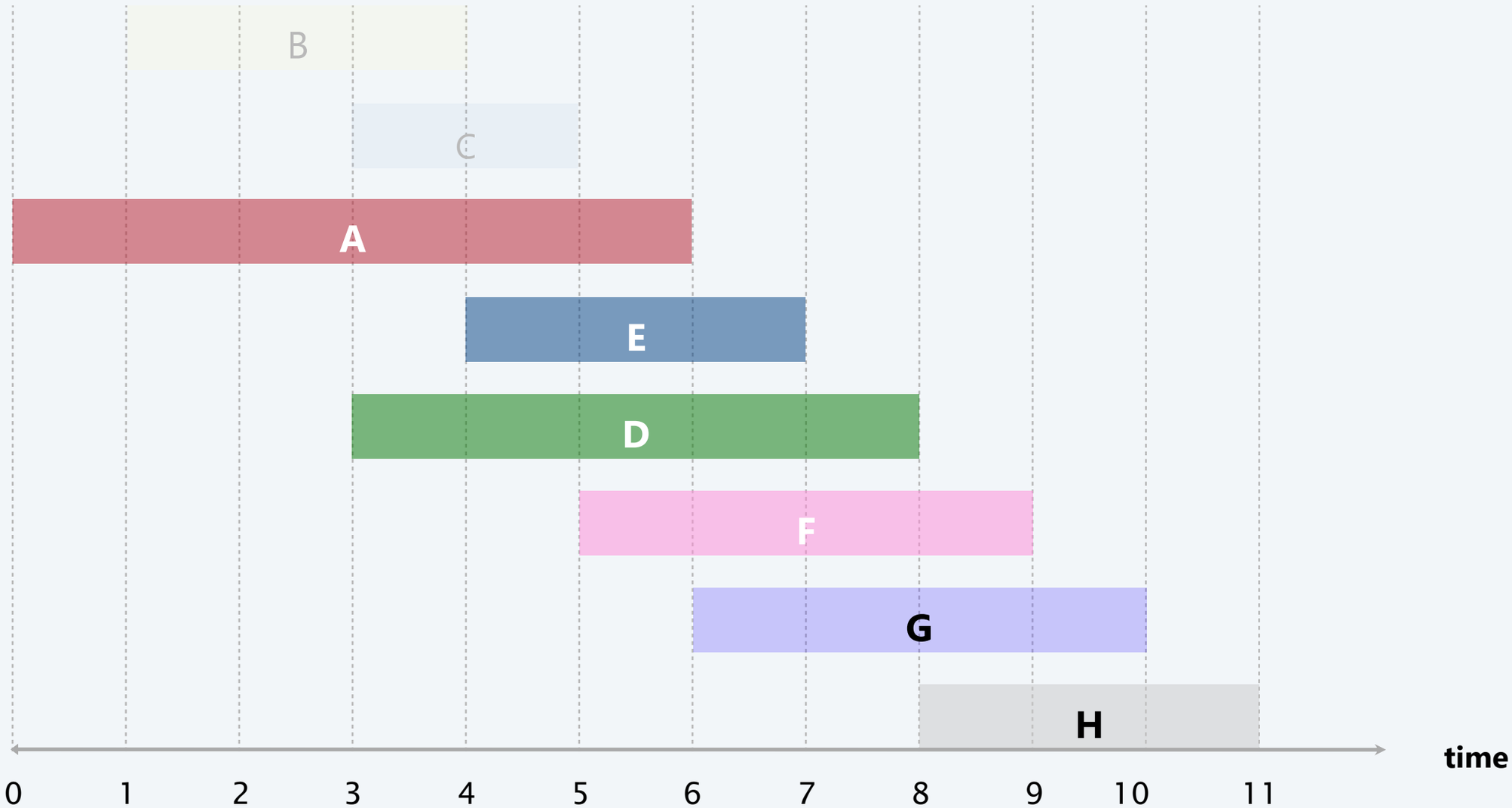
Earliest-finish-time-first algorithm demo

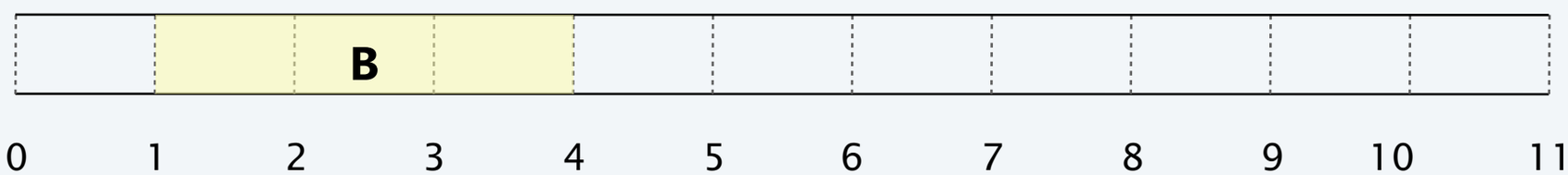


job C is incompatible (do not add to schedule)

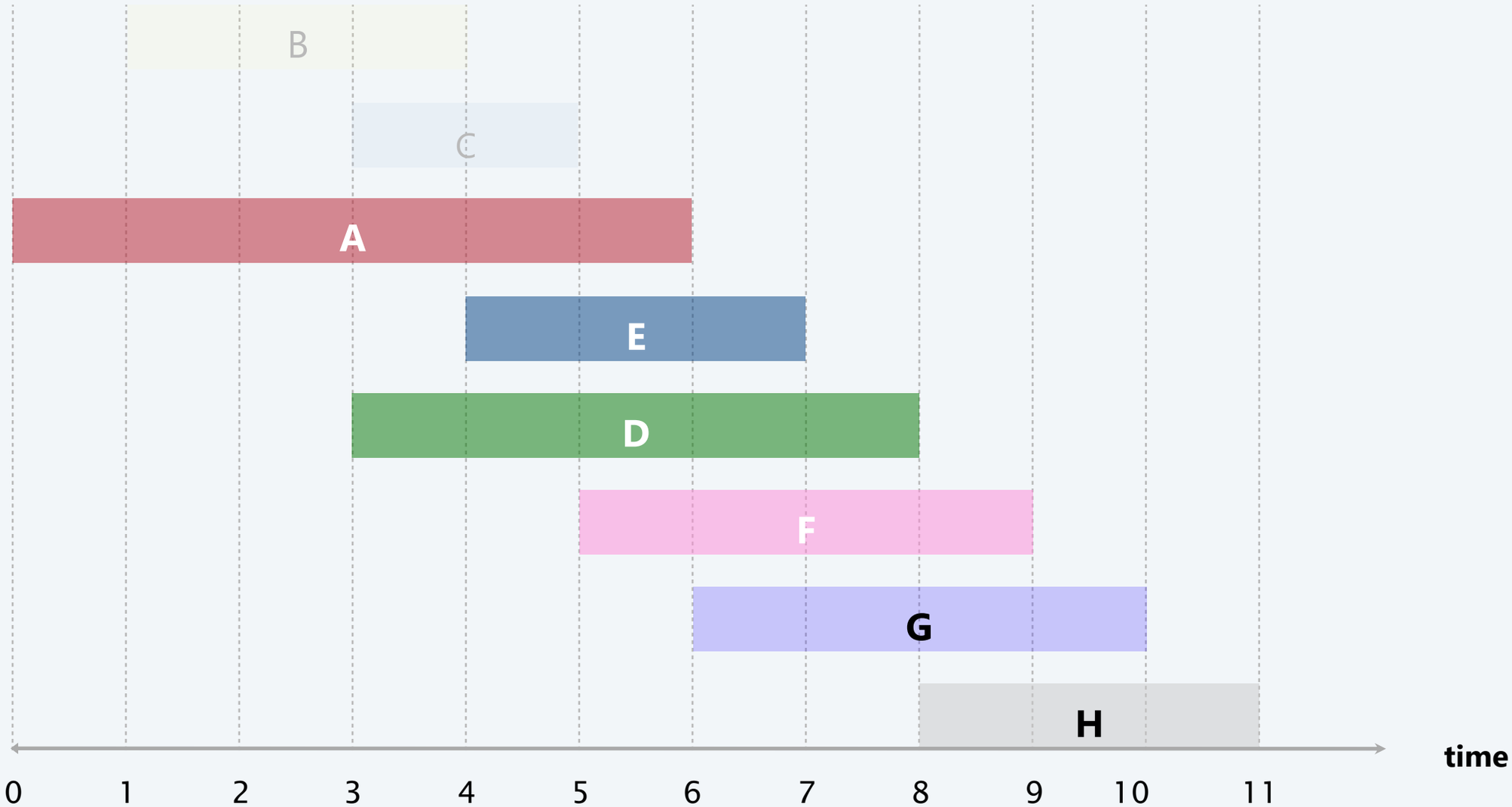


Earliest-finish-time-first algorithm demo

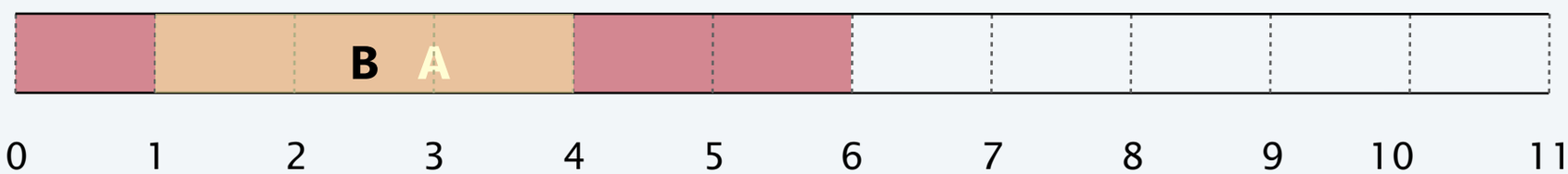




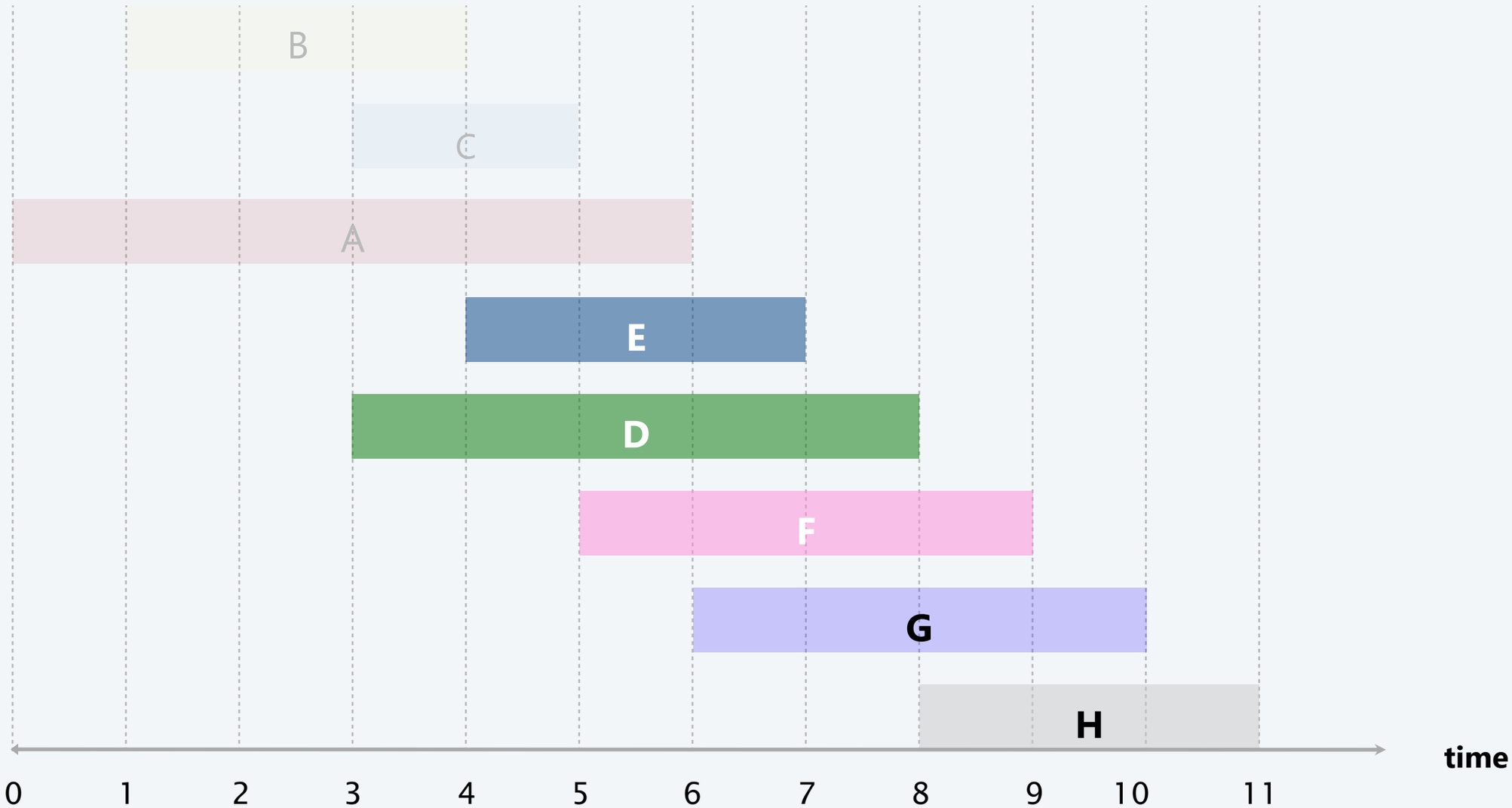
Earliest-finish-time-first algorithm demo

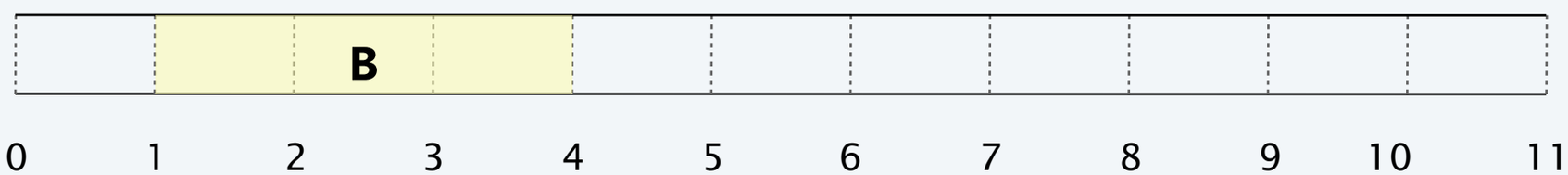


job A is incompatible (do not add to schedule)

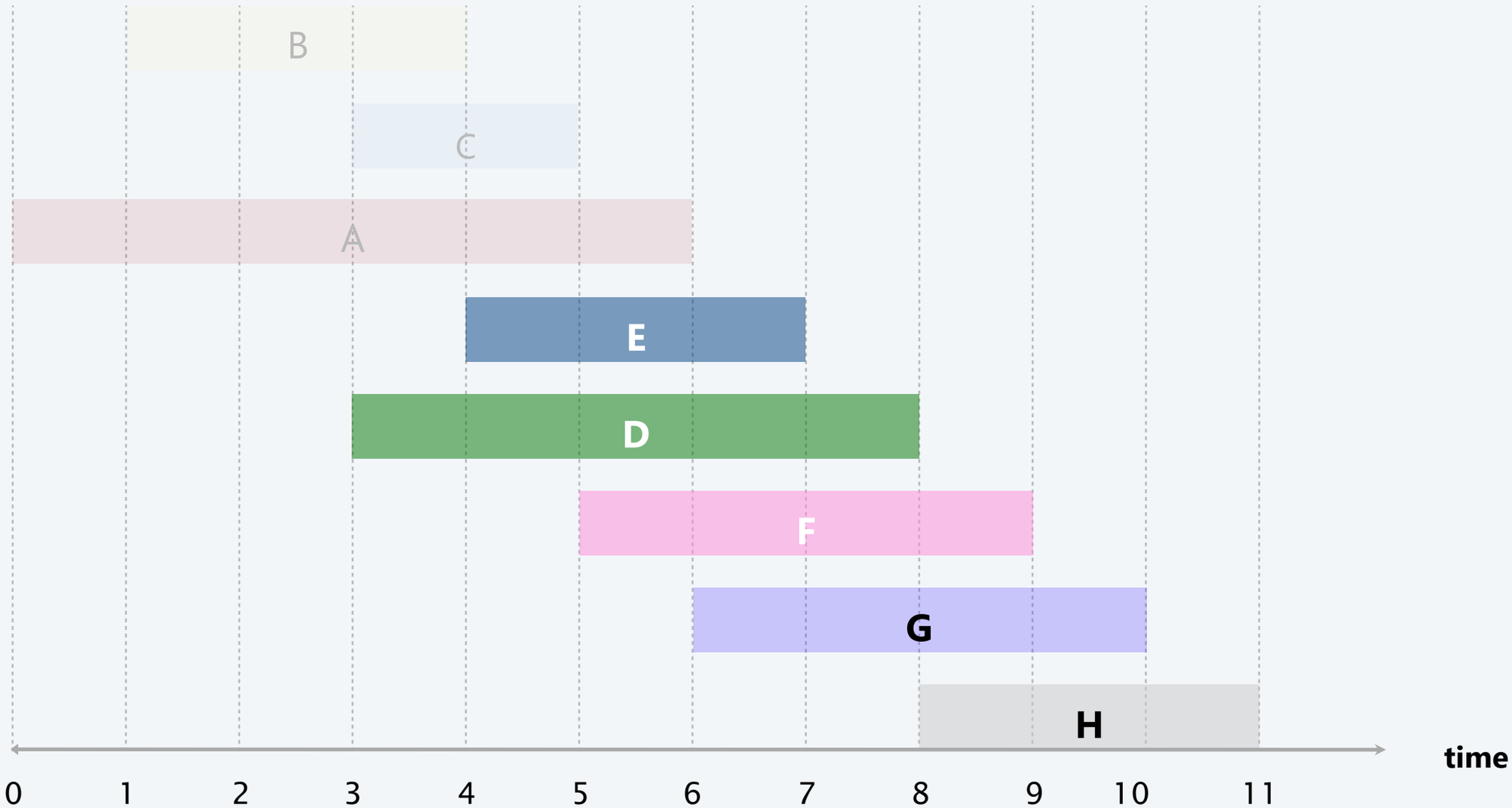


Earliest-finish-time-first algorithm demo

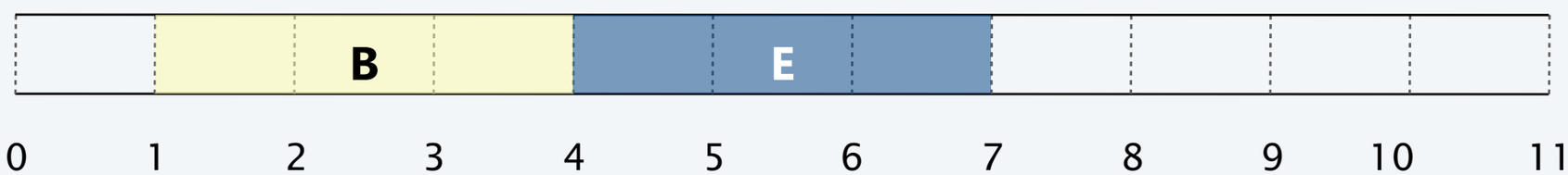




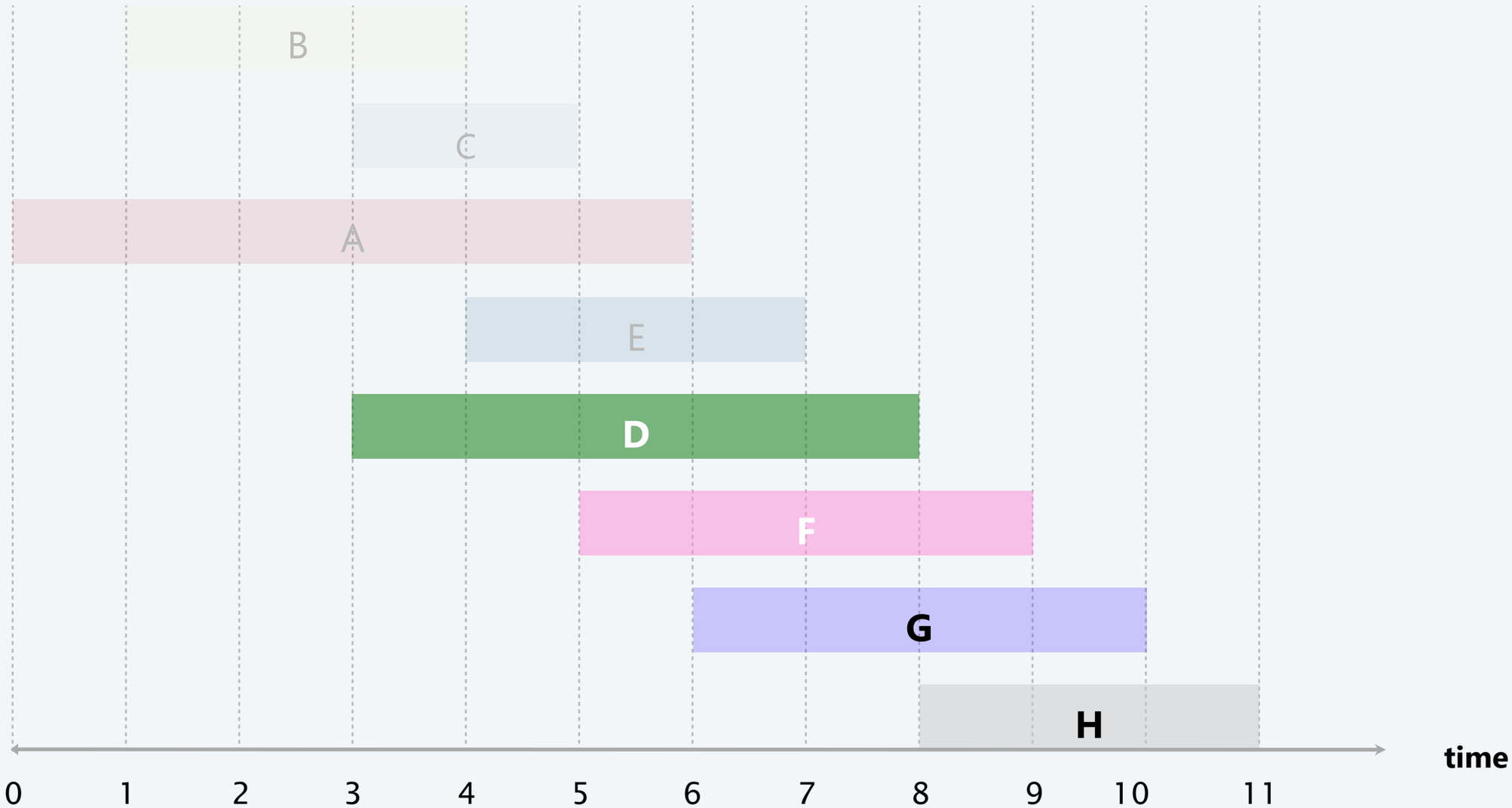
Earliest-finish-time-first algorithm demo

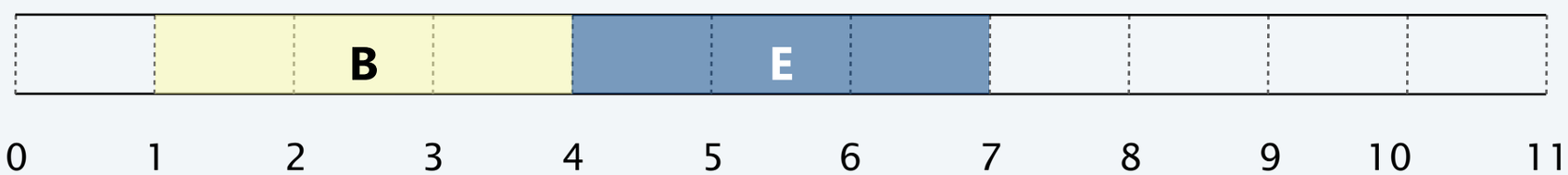


job E is compatible (add to schedule)

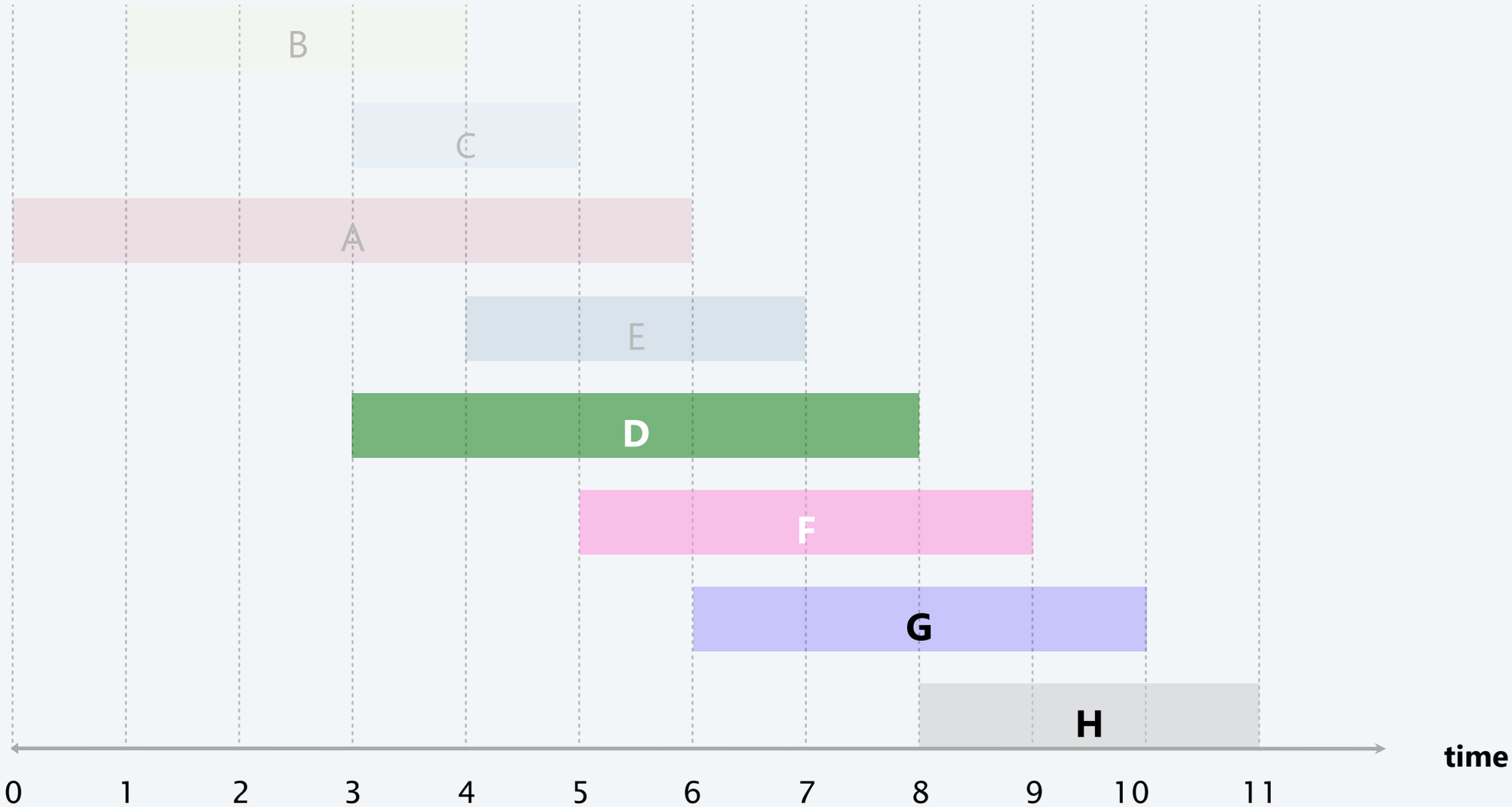


Earliest-finish-time-first algorithm demo

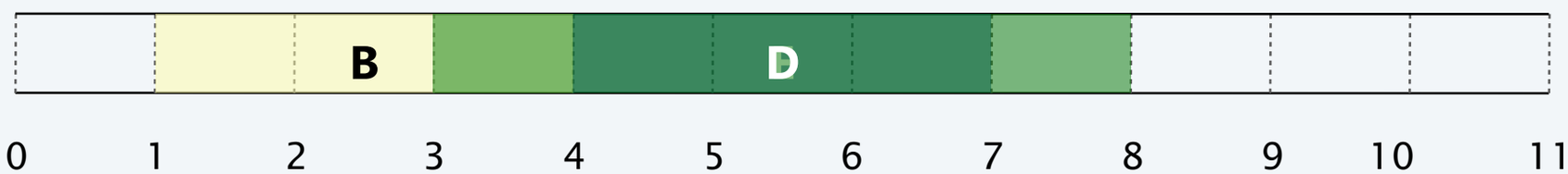




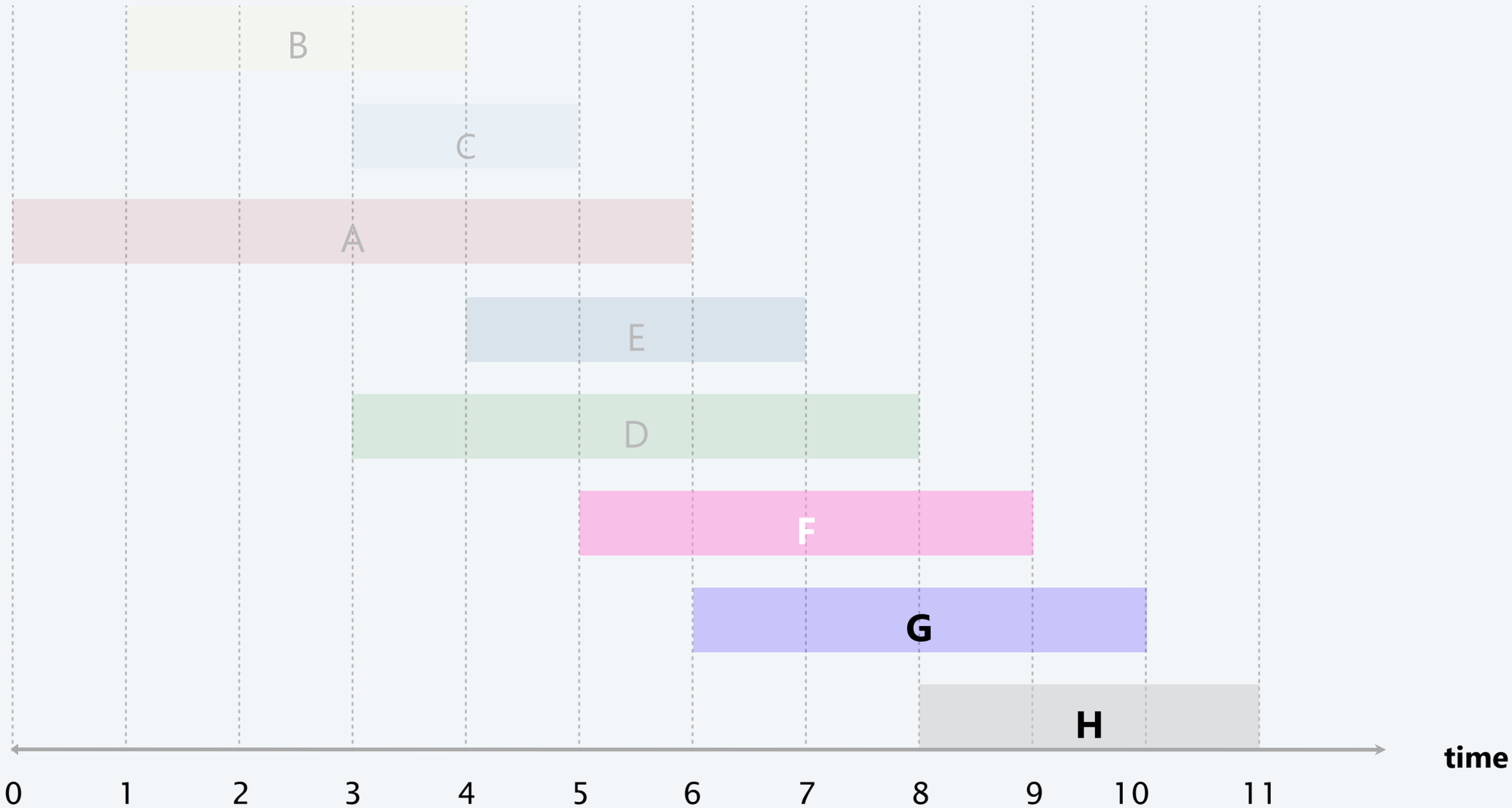
Earliest-finish-time-first algorithm demo

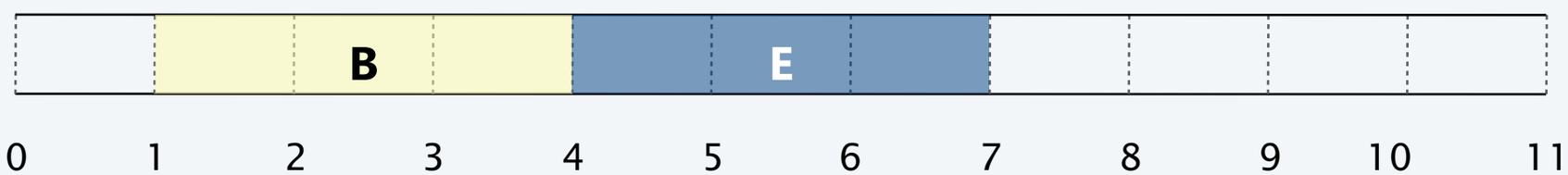


job D is incompatible (do not add to schedule)

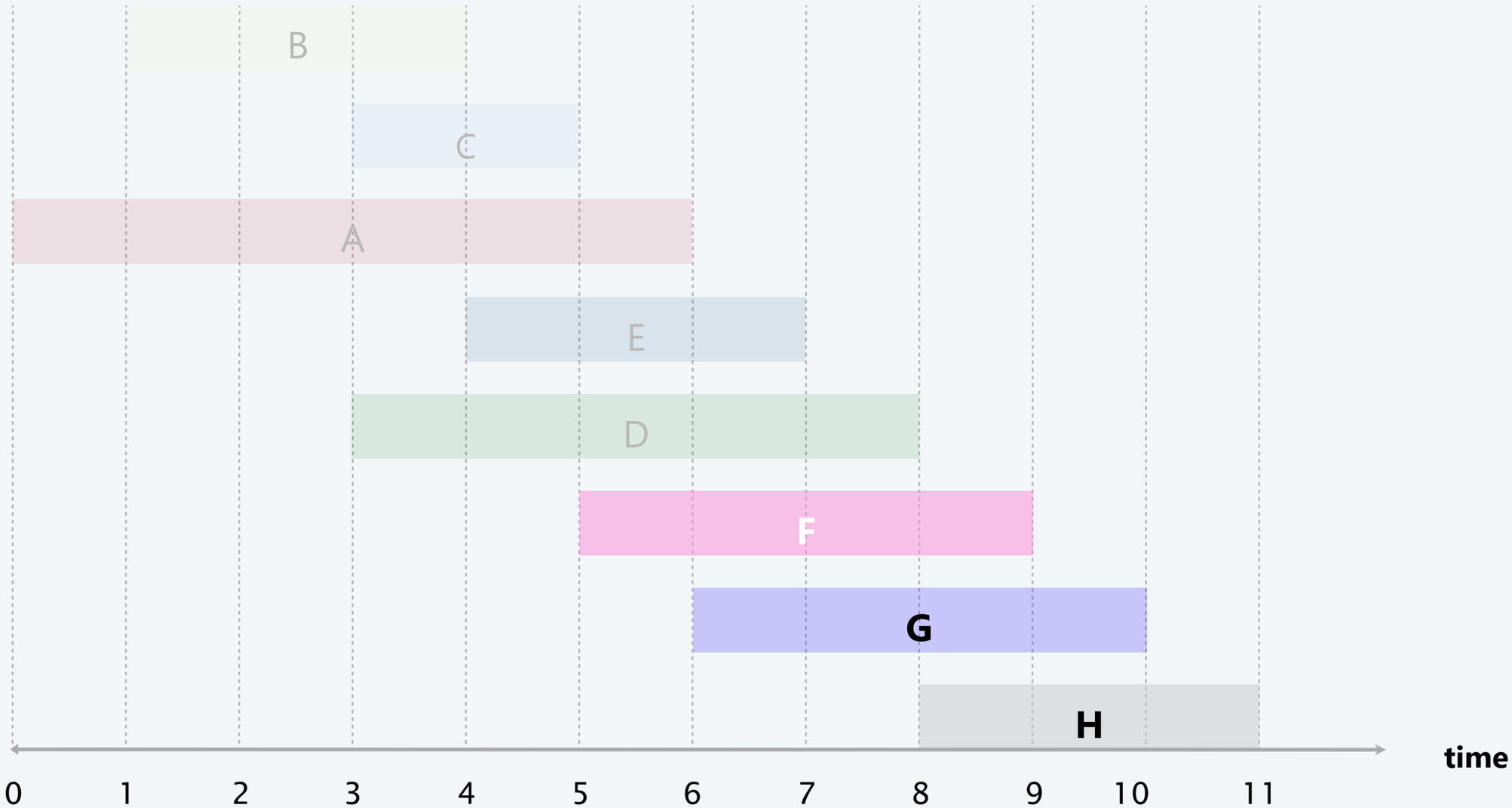


Earliest-finish-time-first algorithm demo

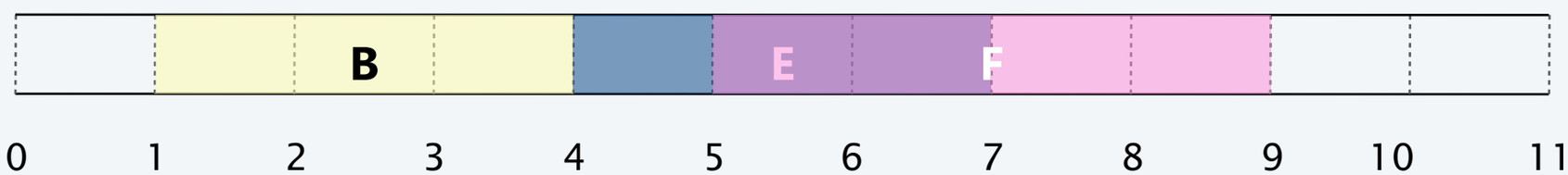




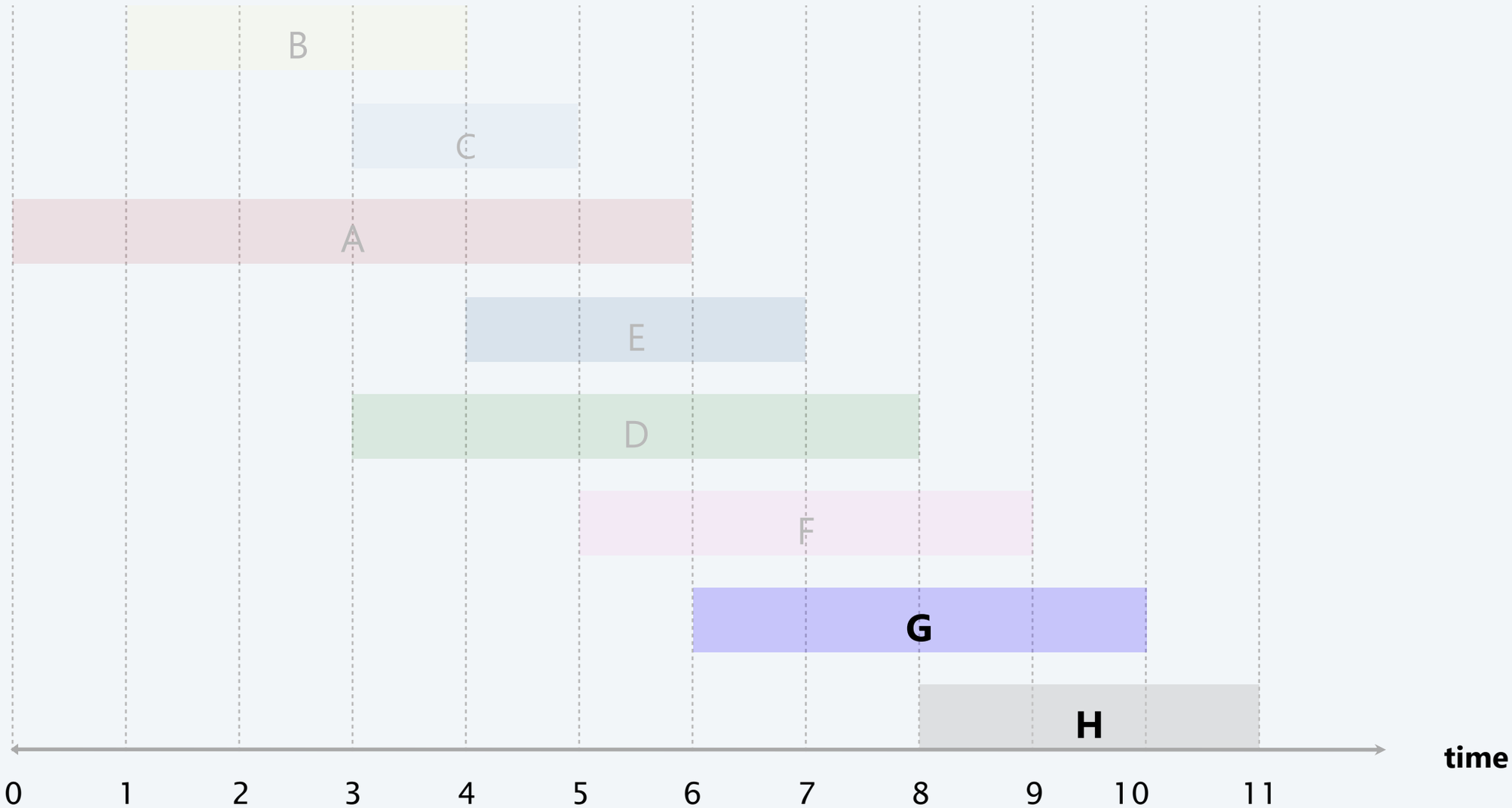
Earliest-finish-time-first algorithm demo

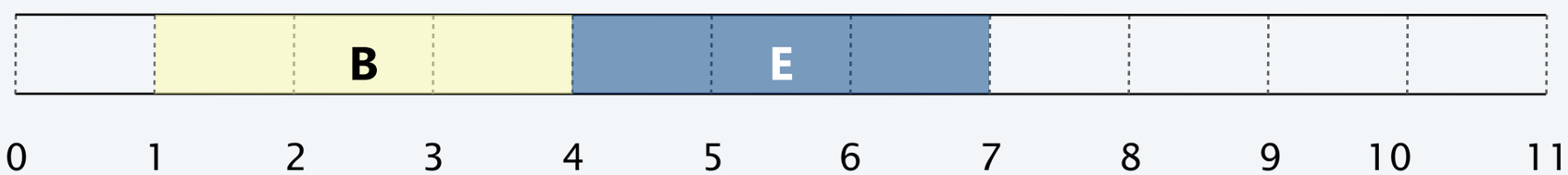


job F is incompatible (do not add to schedule)

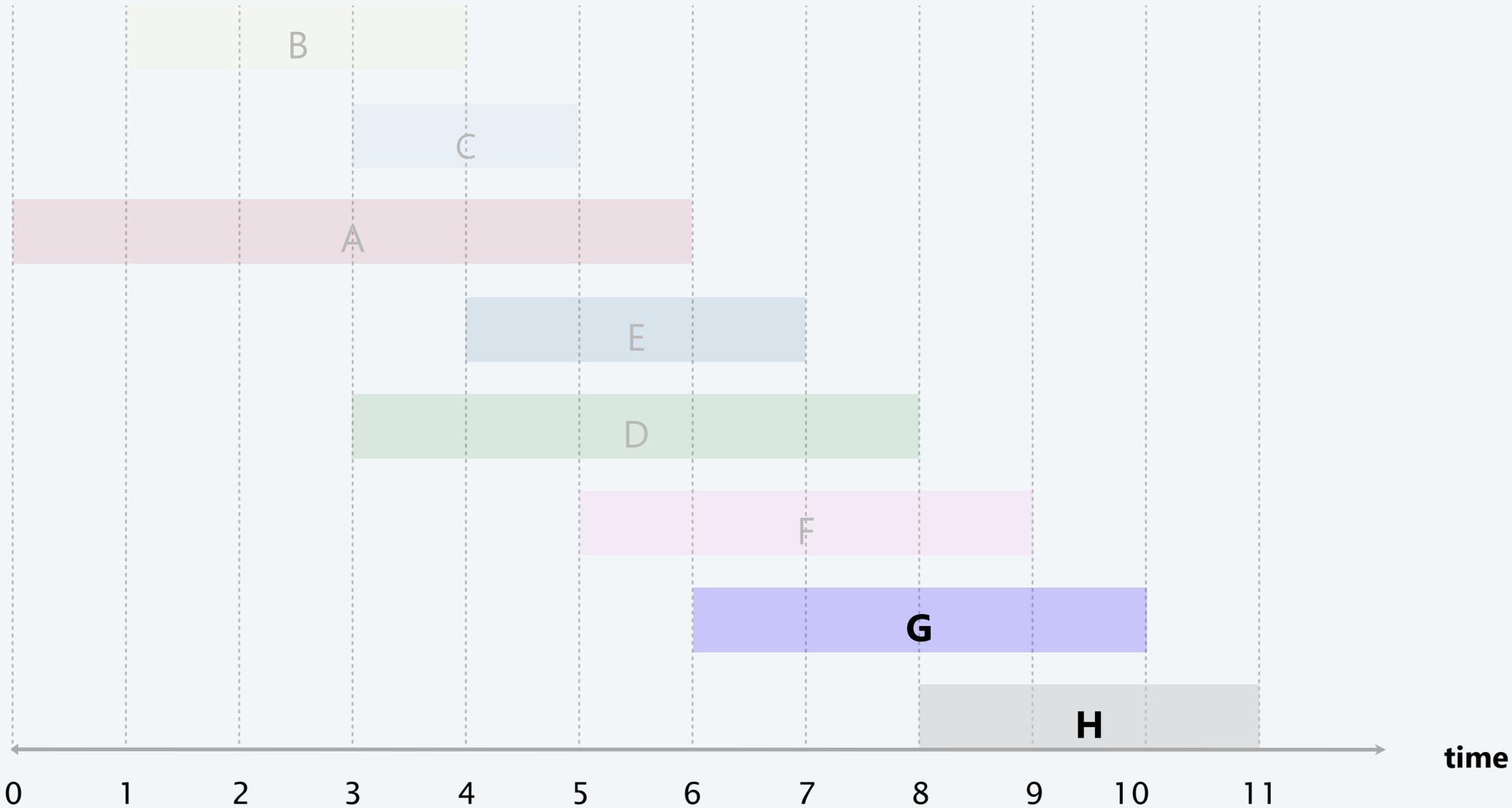


Earliest-finish-time-first algorithm demo

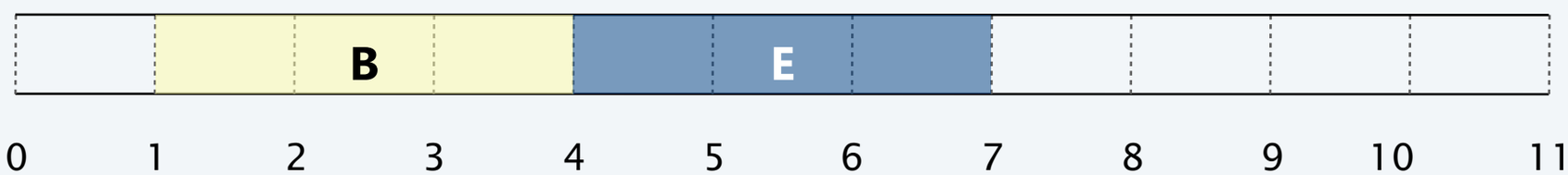




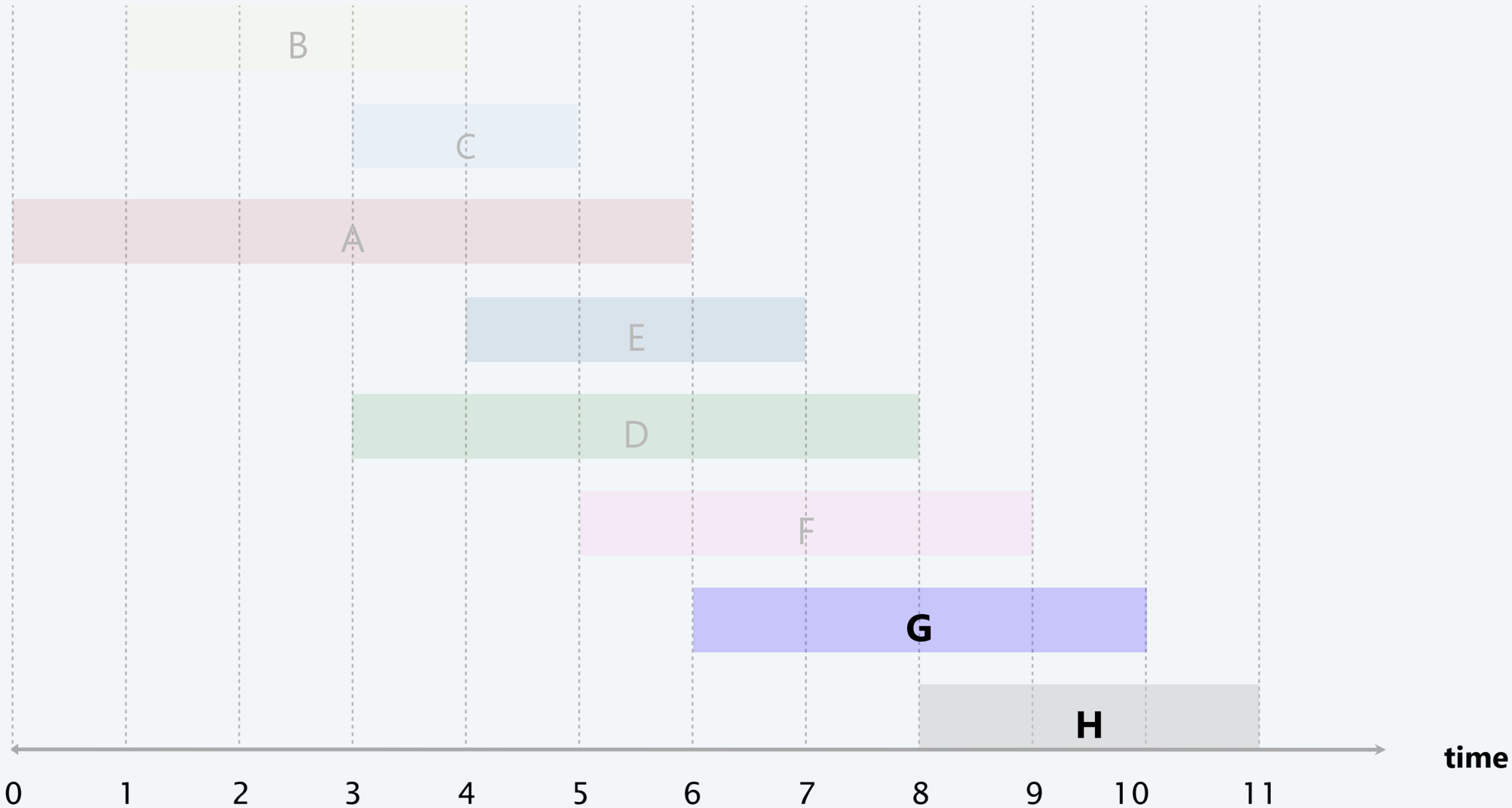
Earliest-finish-time-first algorithm demo



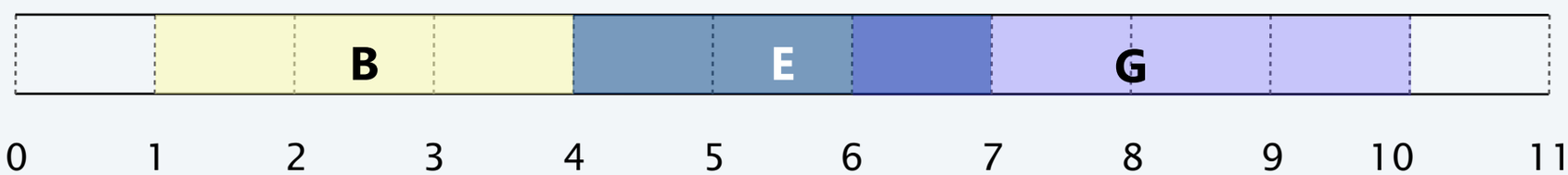
job G is incompatible (do not add to schedule)



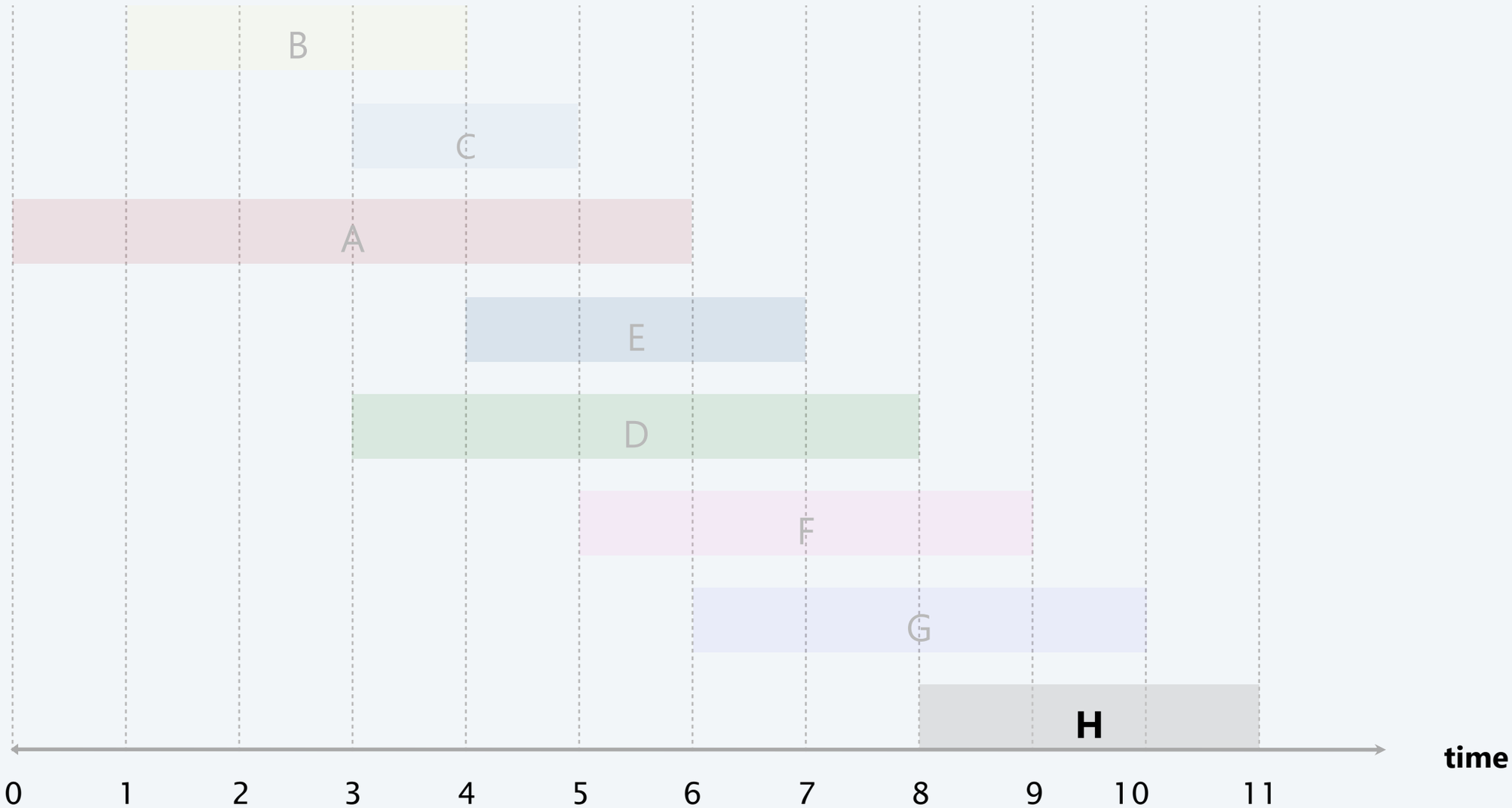
Earliest-finish-time-first algorithm demo

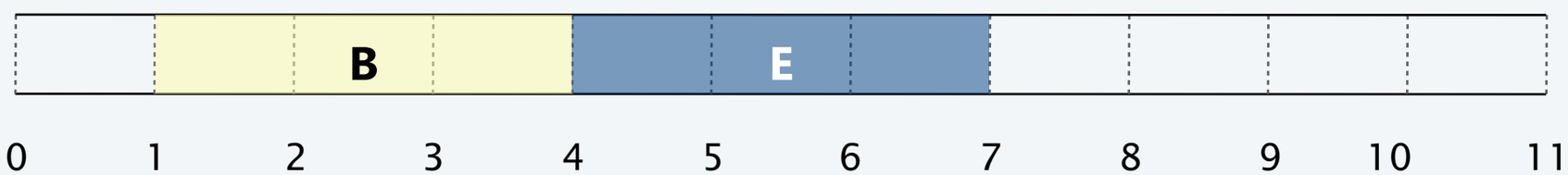


job G is incompatible (do not add to schedule)

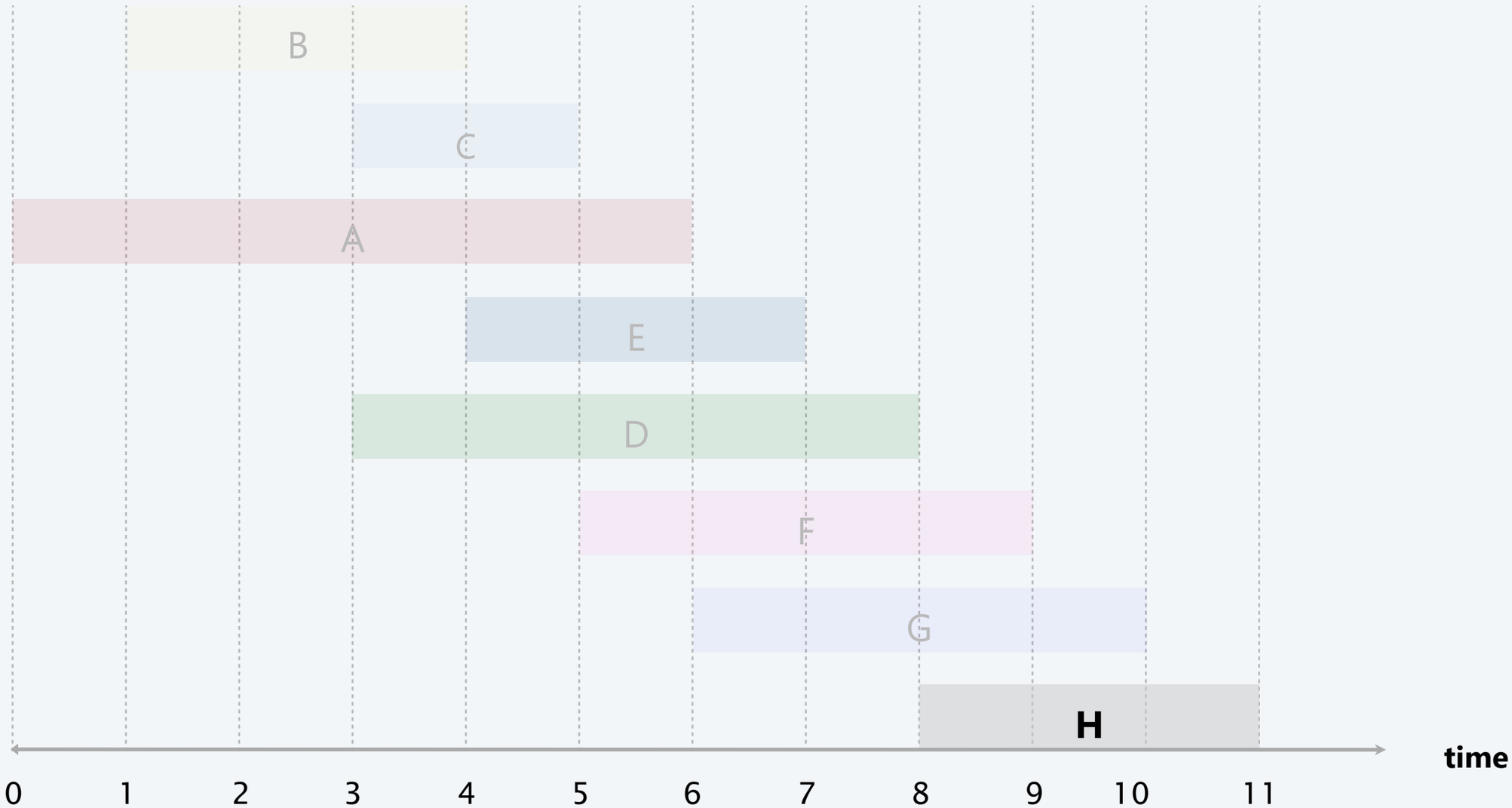


Earliest-finish-time-first algorithm demo

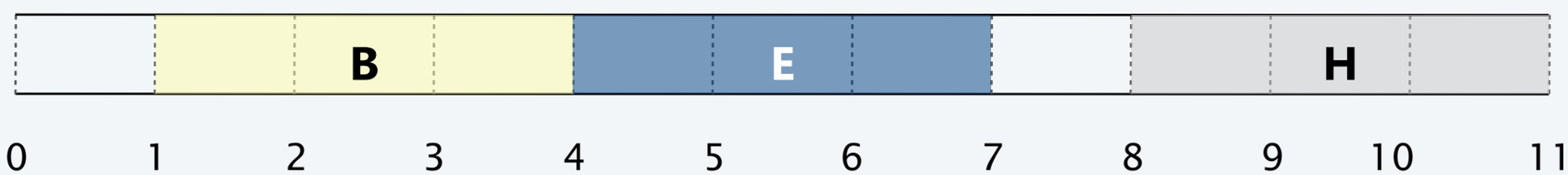




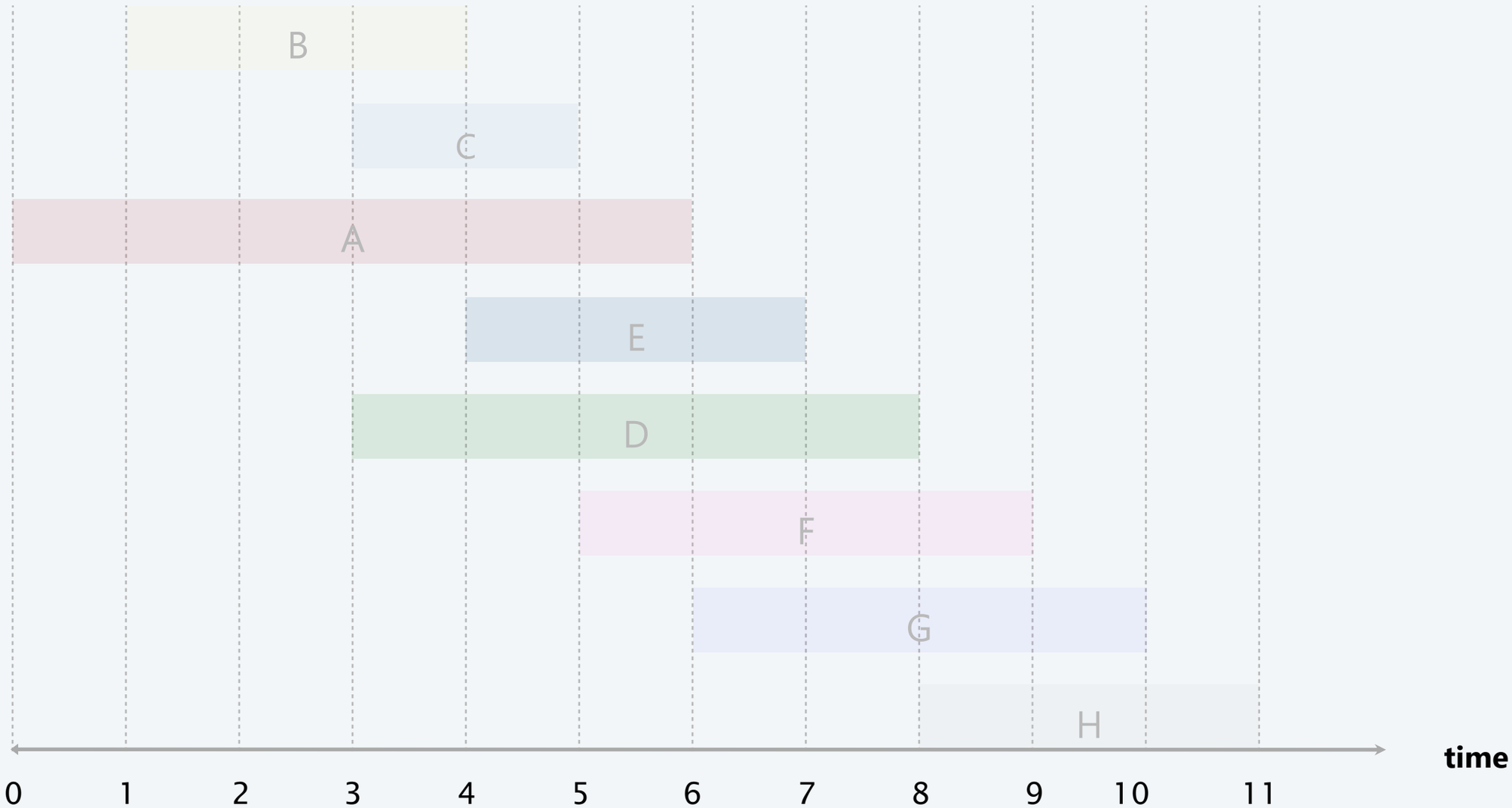
Earliest-finish-time-first algorithm demo

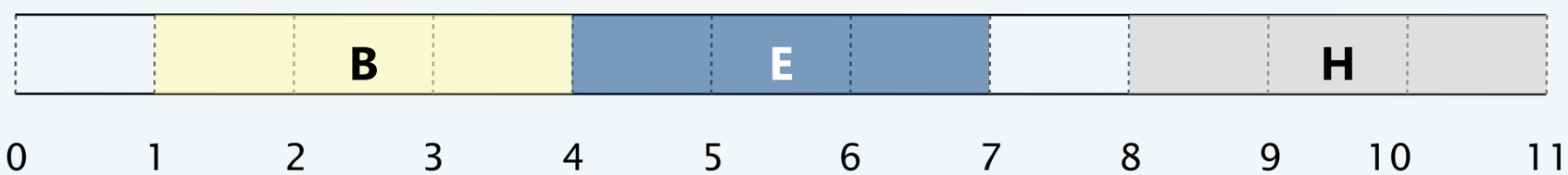


job H is compatible (add to schedule)

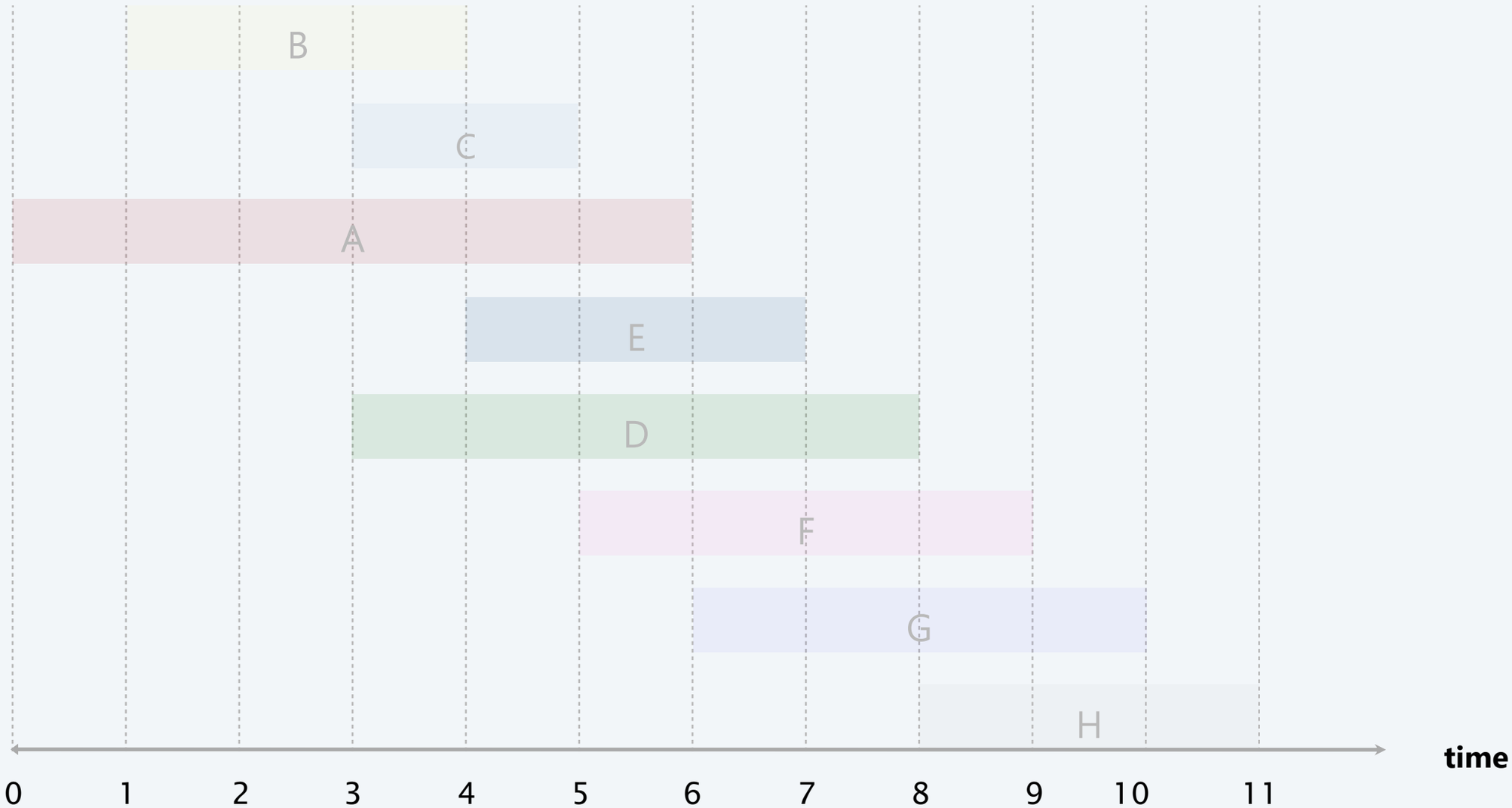


Earliest-finish-time-first algorithm demo

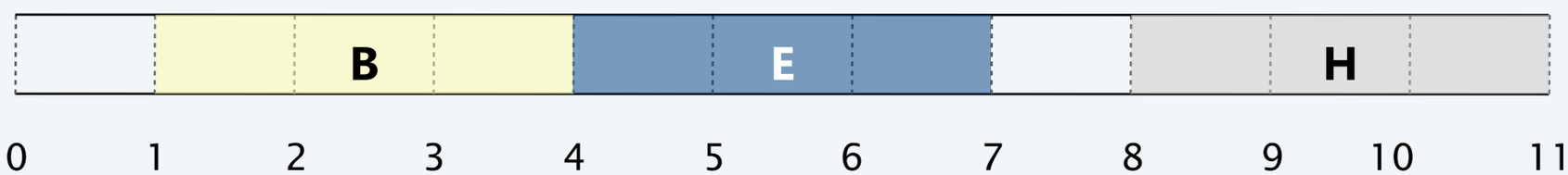




Earliest-finish-time-first algorithm demo



done (an optimal set of jobs)



Interval scheduling: earliest-finish-time-first algorithm

EARLIEST-FINISH-TIME-FIRST $(n, s_1, s_2, \ldots, s_n, f_1, f_2, \ldots, f_n)$

SORT jobs by finish times and renumber so that $f_1 \leq f_2 \leq \ldots \leq f_n$. $\}$ $n \log(n)$

$S \leftarrow \varnothing$. ← set of jobs selected

FOR $j = 1$ TO n

  IF (job j is compatible with S) $\}$ n

    $S \leftarrow S \cup \{ j \}$.

RETURN S.

Costo $= T(n) = n + n\log(n) = O(n\log(n))$

Proposition. Can implement earliest-finish-time first in $O(n \log n)$ time.

- Keep track of job j^* that was added last to S.
- Job j is compatible with S iff $s_j \geq f_{j^*}$.
- Sorting by finish times takes $O(n \log n)$ time.

se il job attuale INIZIA dopo la FINE del job precedente va bene

(j^*)

Interval scheduling: analysis of earliest-finish-time-first algorithm

Let $i_1, i_2, \ldots i_k$ be set of jobs selected by greedy (ordered by finish times).
Let $j_1, j_2, \ldots j_m$ be set of jobs in an optimal solution (ordered by finish times)

denote by $f(i_r)$ the finish time of job i_r .

condizione

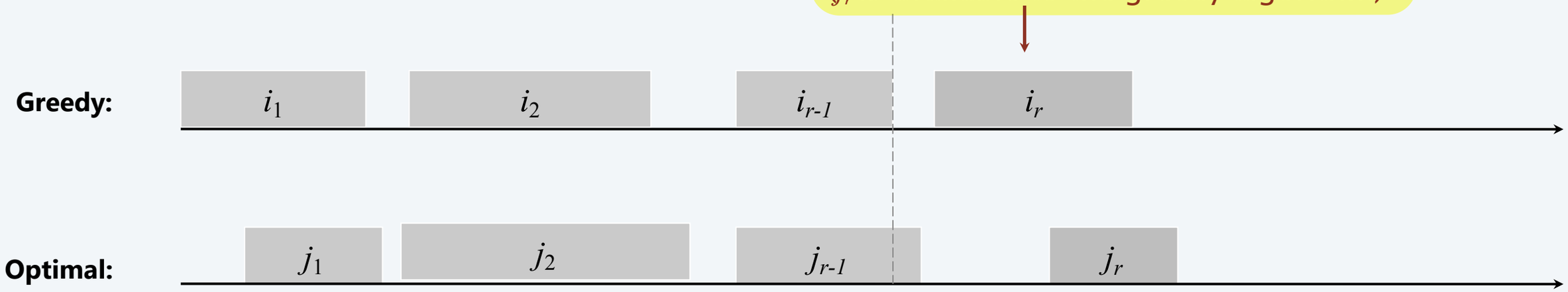
Lemma. For every r=1,2...,k, we have $f(i_r) \leq f(j_r)$.

N.B. (*) = Sia il greedy che l'ottimale DEVONO avere lo stesso numero di job presi

Pf. [by induction]

- r=1: Obvious.
- r>1:

job i_r must finish no later than j_r
(j_r is available to the greedy algorithm)

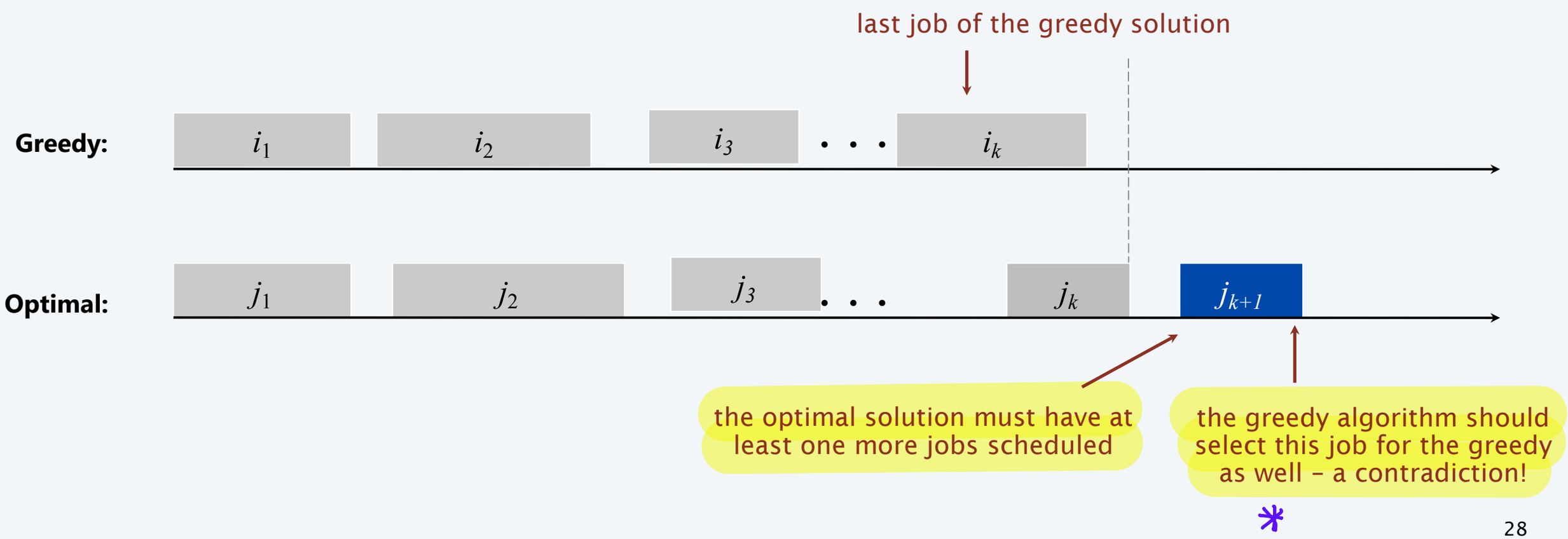


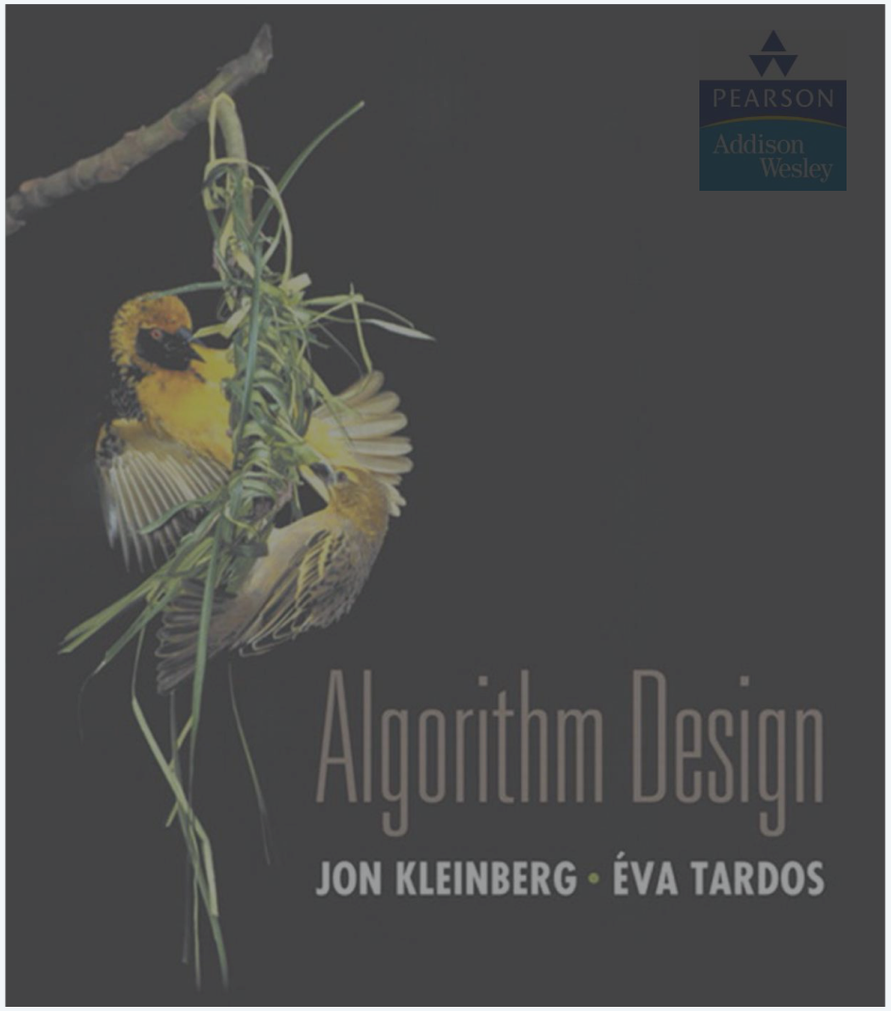
Interval scheduling: analysis of earliest-finish-time-first algorithm

Theorem. The earliest-finish-time-first algorithm is optimal.

Pf. [by contradiction]

- Let i_1, i_2, … i_k be set of jobs selected by greedy (ordered by finish times).
- Let j_1, j_2, … j_m be set of jobs in an optimal solution (ordered by finish times)
- Assume greedy is not optimal
  - then $m>k$





SECTION 4.1

4. GREEDY ALGORITHMS I

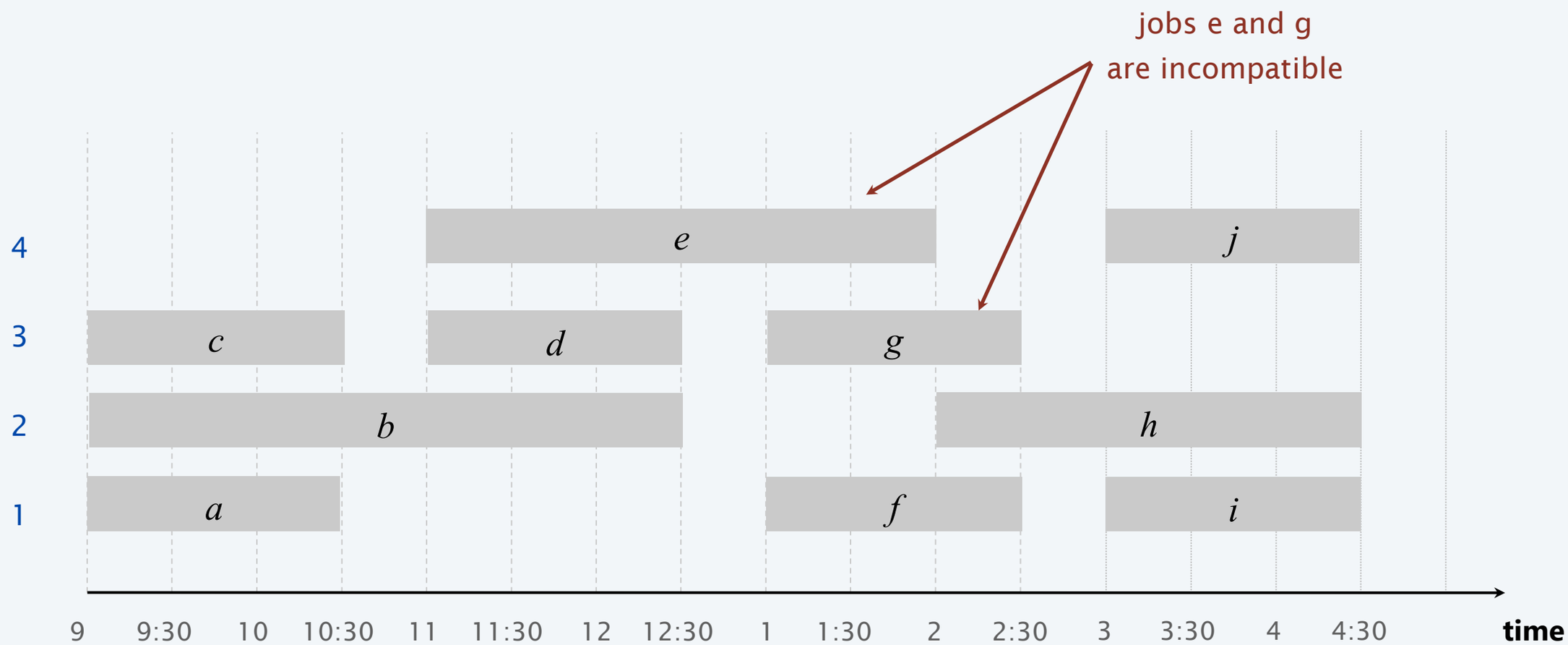
A related problem:

- *interval partitioning*

Interval partitioning

- Lecture j starts at s_j and finishes at f_j.
- Goal: find minimum number of classrooms to schedule all lectures so that no two lectures occur at the same time in the same room.

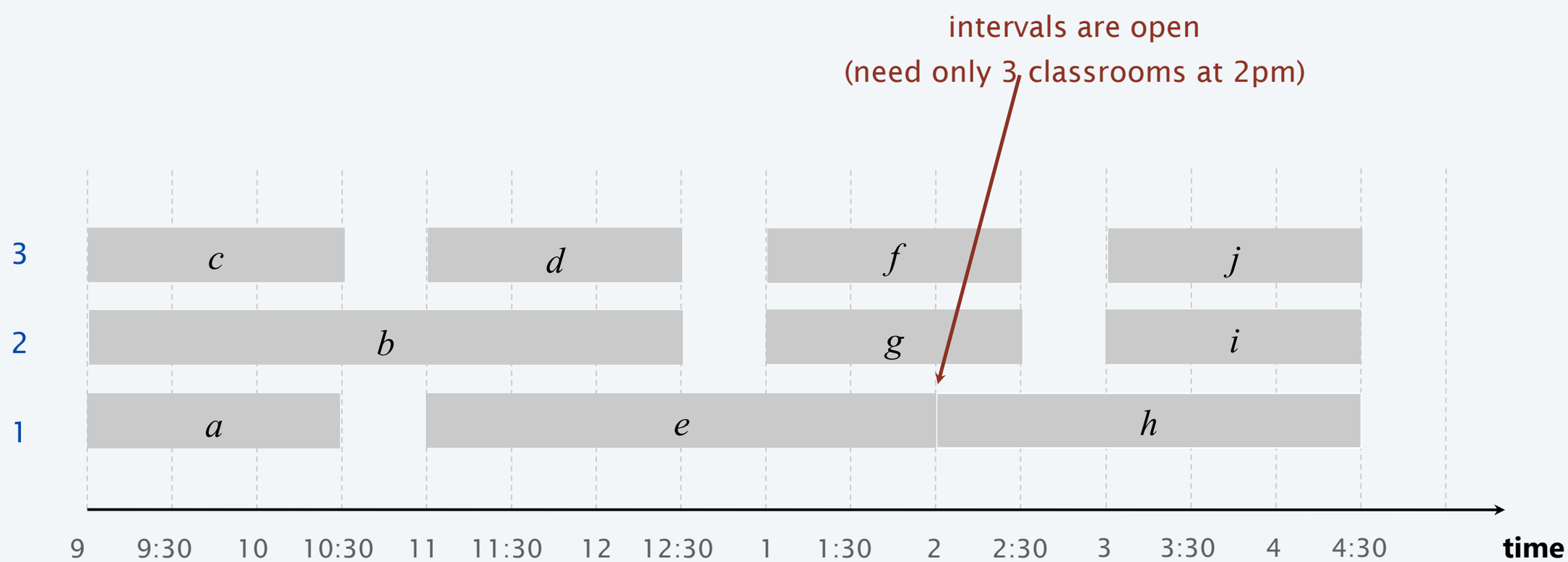
Ex. This schedule uses 4 classrooms to schedule 10 lectures.



Interval partitioning

- Lecture j starts at s_j and finishes at f_j.
- Goal: find minimum number of classrooms to schedule all lectures so that no two lectures occur at the same time in the same room.

Ex. This schedule uses 3 classrooms to schedule 10 lectures.



Interval partitioning

- Input:
  - A set of n intervals $I_1,\ldots,I_n$
  - interval I_i has starting time s_i and finish time f_i
- Feasible solution:
  - A partition of the intervals into subsets (called classrooms) $C_1,\ldots,C_d$ such that each C_i contains mutually compatible intervals
- Measure (to minimize):
  - number of classrooms, i.e. d

Interval partitioning: greedy algorithms

Greedy template. Consider lectures in some natural order. Assign each lecture to an available classroom (which one?); allocate a new classroom if none are available.

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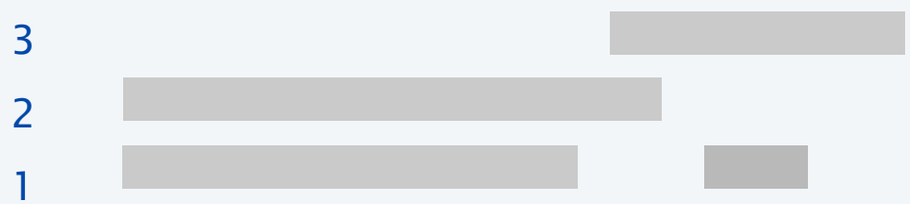
- [Earliest start time] Consider lectures in ascending order of s_j.
- [Earliest finish time] Consider lectures in ascending order of f_j.
- [Shortest interval] Consider lectures in ascending order of $f_j - s_j$.
- [Fewest conflicts] For each lecture j, count the number of conflicting lectures c_j. Schedule in ascending order of c_j.

Interval partitioning: greedy algorithms

Greedy template. Consider lectures in some natural order. Assign each lecture to an available classroom (which one?); allocate a new classroom if none are available.

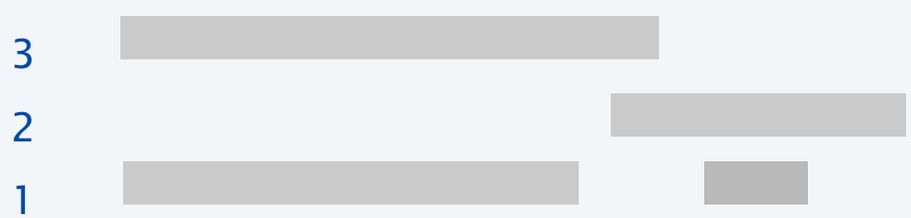
counterexample for earliest finish time



Ne ho usate 3

Potevo usarne 2

counterexample for shortest interval



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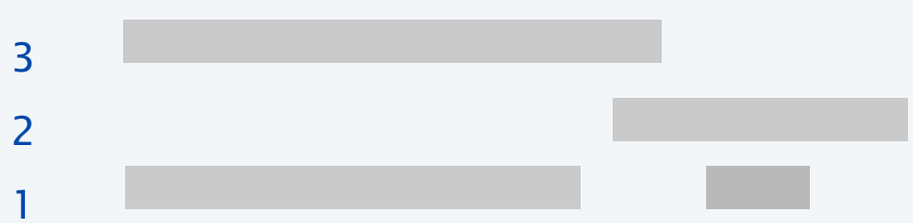
counterexample for fewest conflicts



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Potevo usarne 2

Interval partitioning: earliest-start-time-first algorithm

EARLIEST-START-TIME-FIRST $(n, s_1, s_2, \ldots, s_n, f_1, f_2, \ldots, f_n)$

SORT lectures by start times and renumber so that $s_1 \leq s_2 \leq \ldots \leq s_n$.

$d \leftarrow 0$. ← number of allocated classrooms

FOR $j = 1$ TO n

  IF (lecture j is compatible with some classroom)

    Schedule lecture j in any such classroom k.

  ELSE

    Allocate a new classroom $d + 1$.

    Schedule lecture j in classroom $d + 1$.

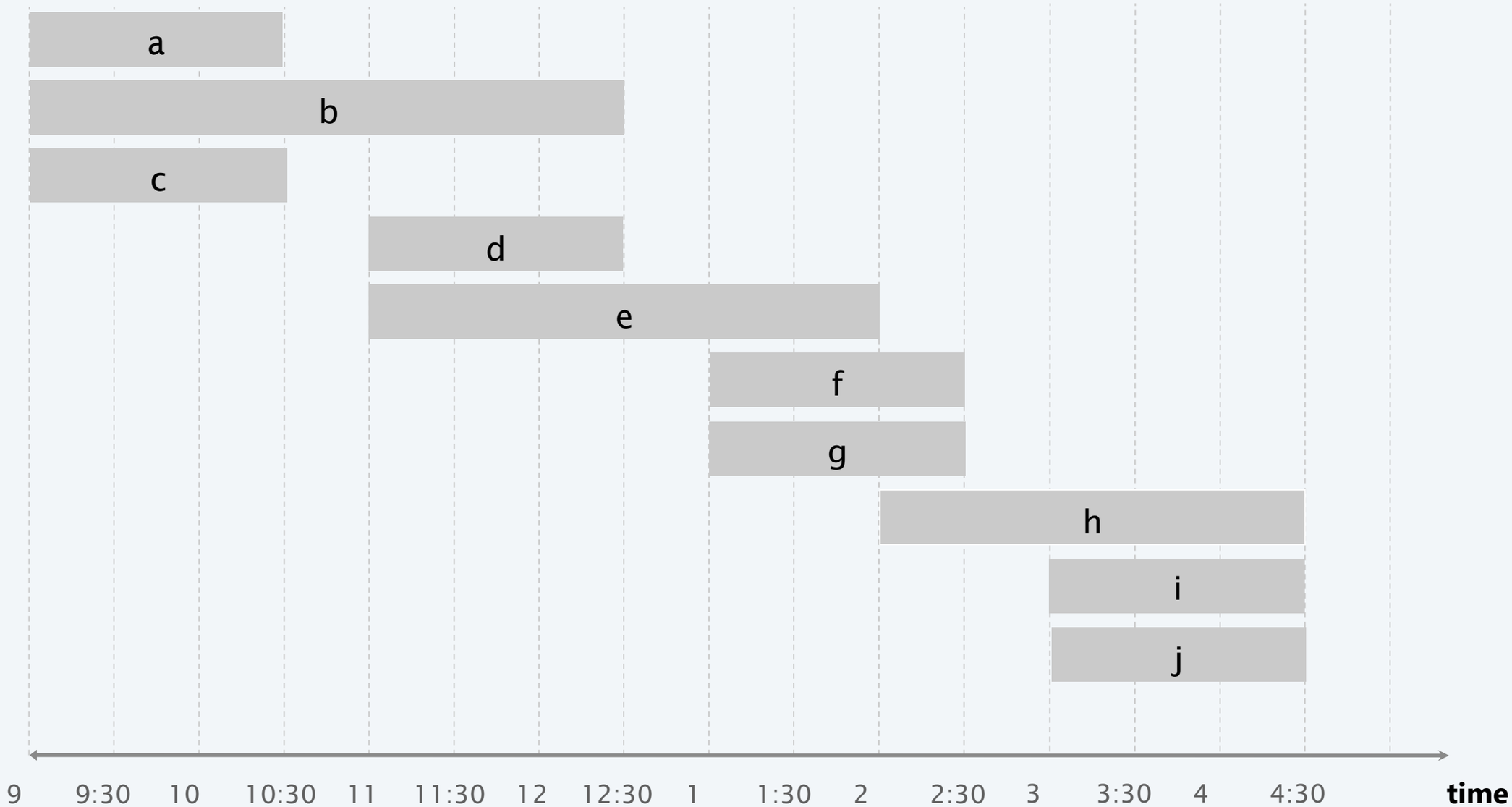
    $d \leftarrow d + 1$.

RETURN schedule.

Earliest-start-time-first algorithm demo

Consider lectures in order of start time:

- Assign next lecture to any compatible classroom (if one exists).
- Otherwise, open up a new classroom.



Earliest-start-time-first algorithm demo

Consider lectures in order of start time:

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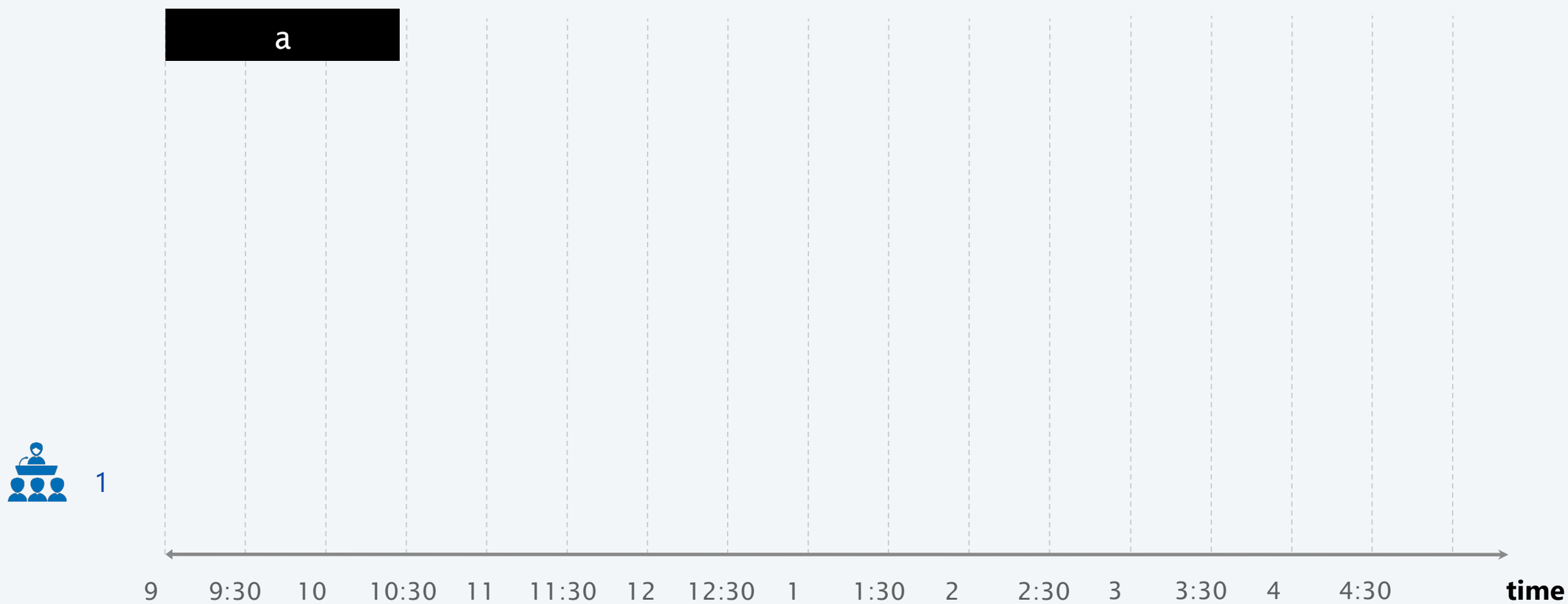
- Otherwise, open up a new classroom.

no compatible classroom: open up a new classroom and assign lecture to it



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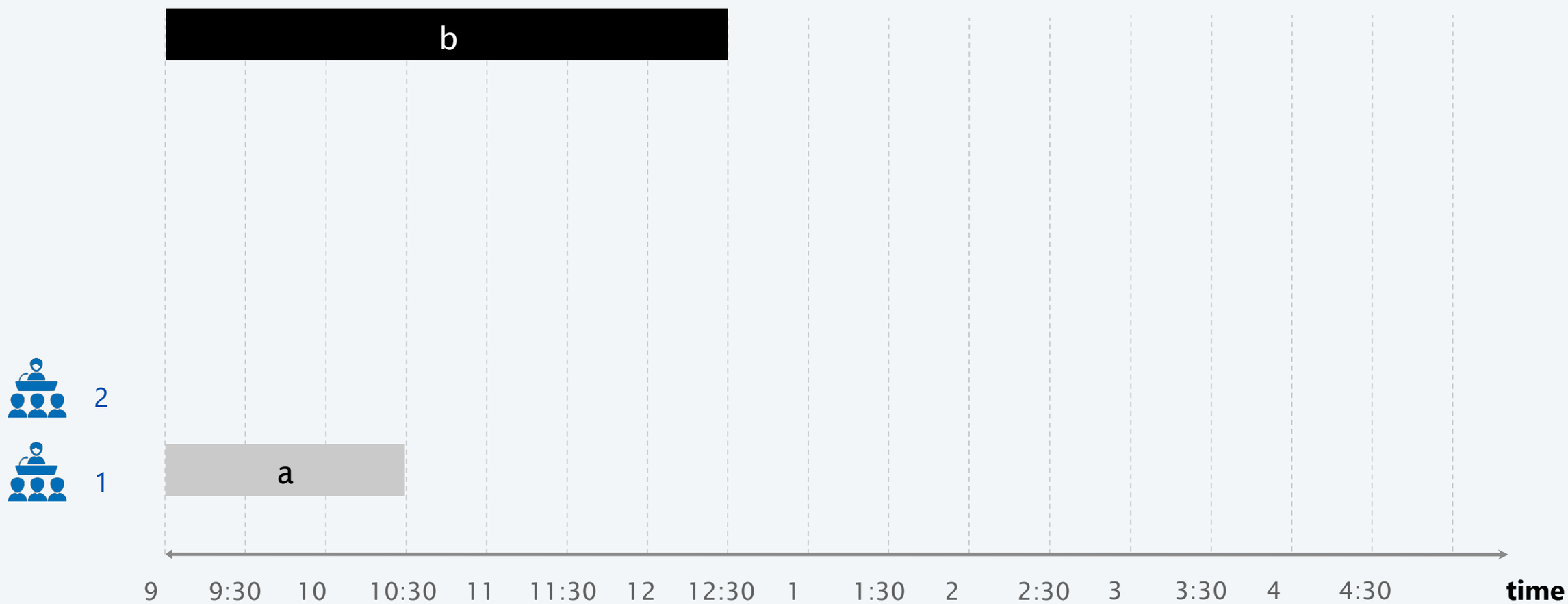

Earliest-start-time-first algorithm demo

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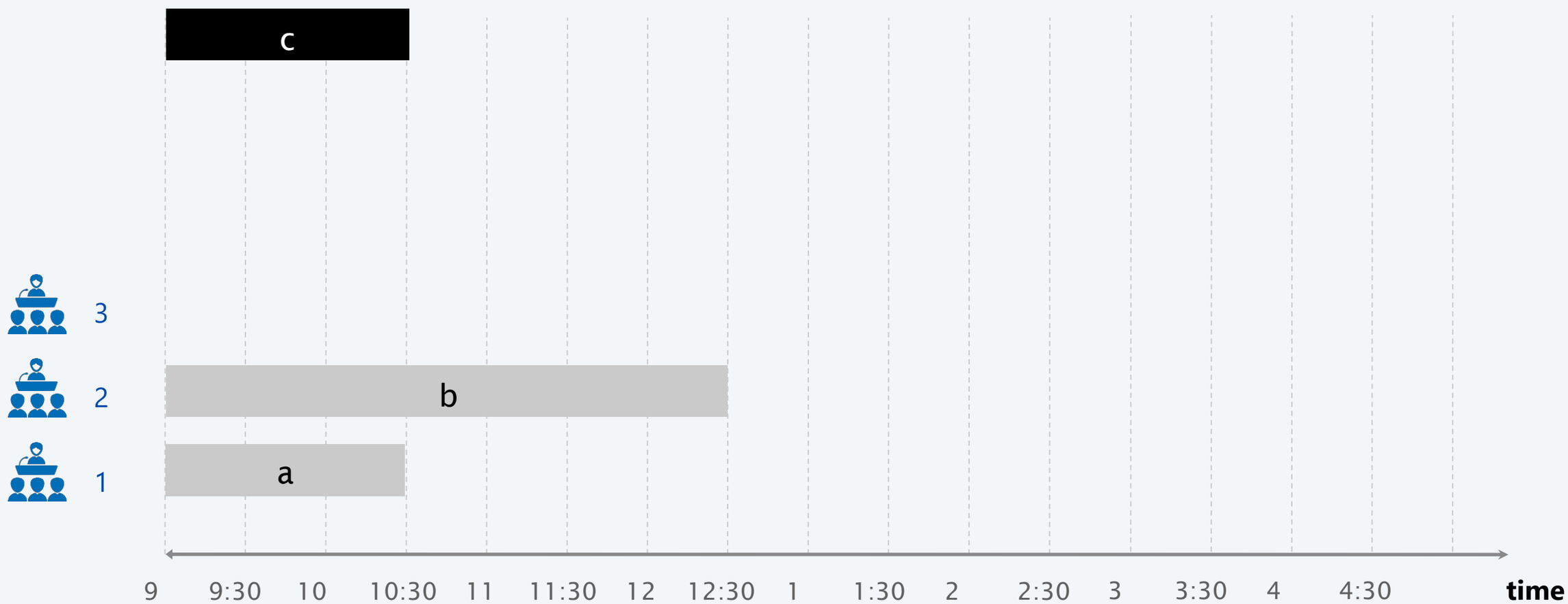
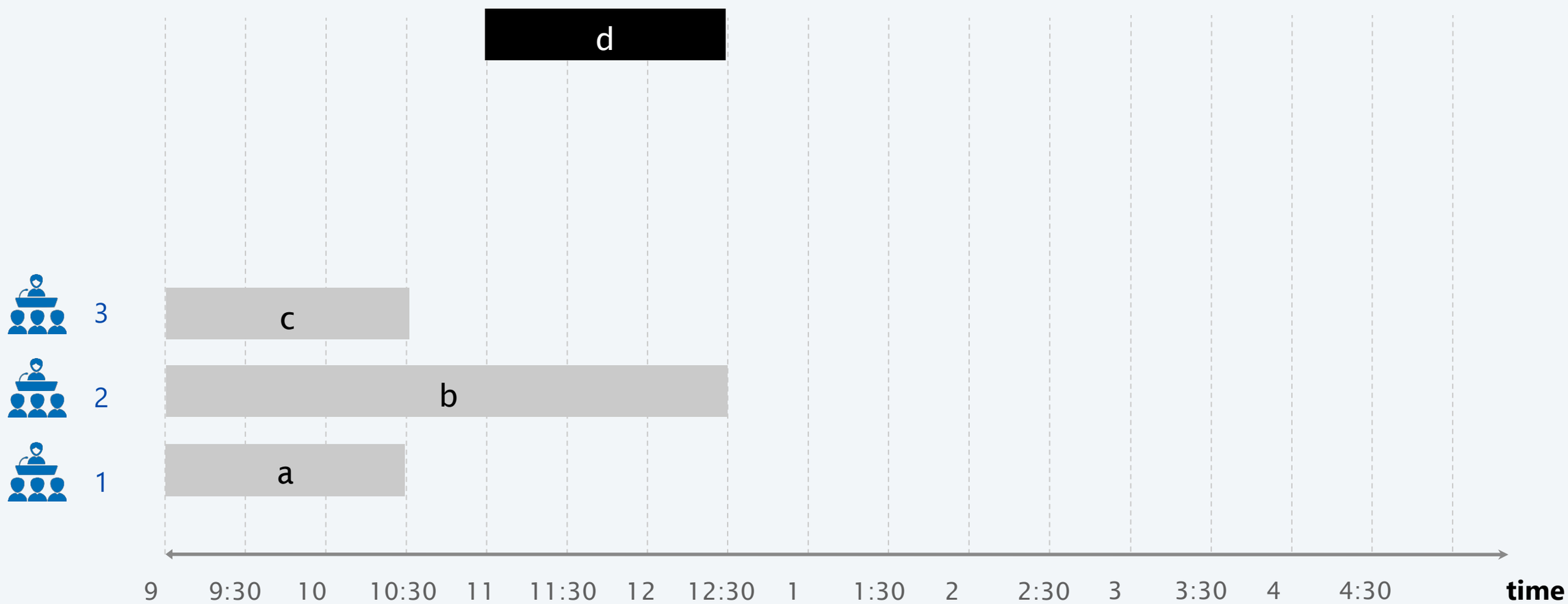


Earliest-start-time-first algorithm demo

Consider lectures in order of start time:

- Assign next lecture to any compatible classroom (if one exists).
- Otherwise, open up a new classroom.

lecture d is compatible with classrooms 1 and 3

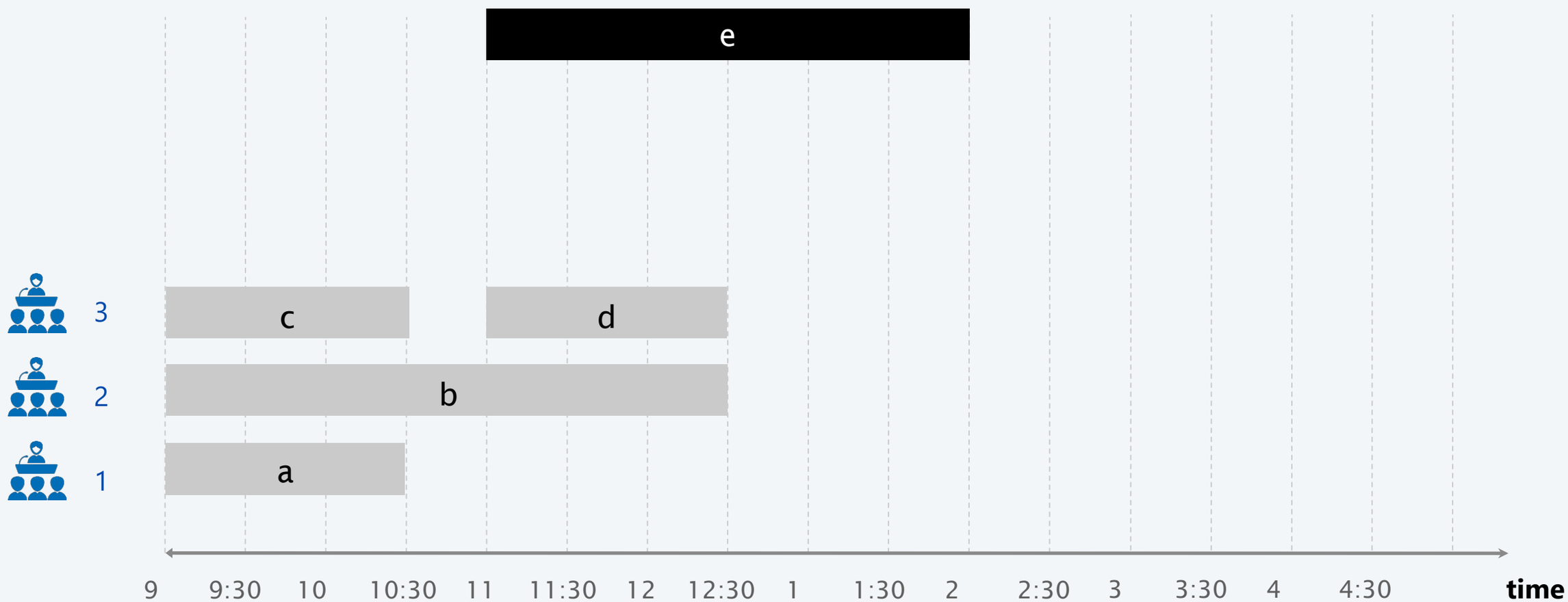


Earliest-start-time-first algorithm demo

Consider lectures in order of start time:

- Assign next lecture to any compatible classroom (if one exists).
- Otherwise, open up a new classroom.

lecture e is compatible with classroom 1

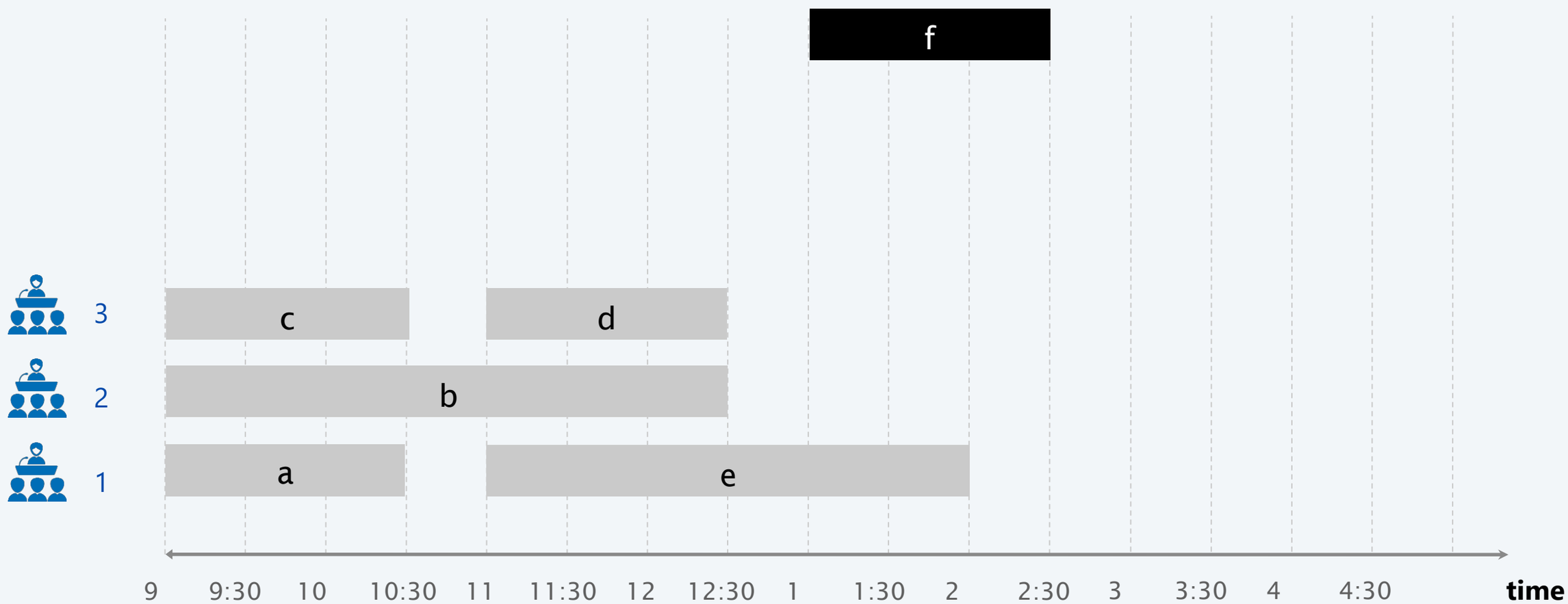


Earliest-start-time-first algorithm demo

Consider lectures in order of start time:

- Assign next lecture to any compatible classroom (if one exists).
- Otherwise, open up a new classroom.

lecture f is compatible with classroom 2 and 3

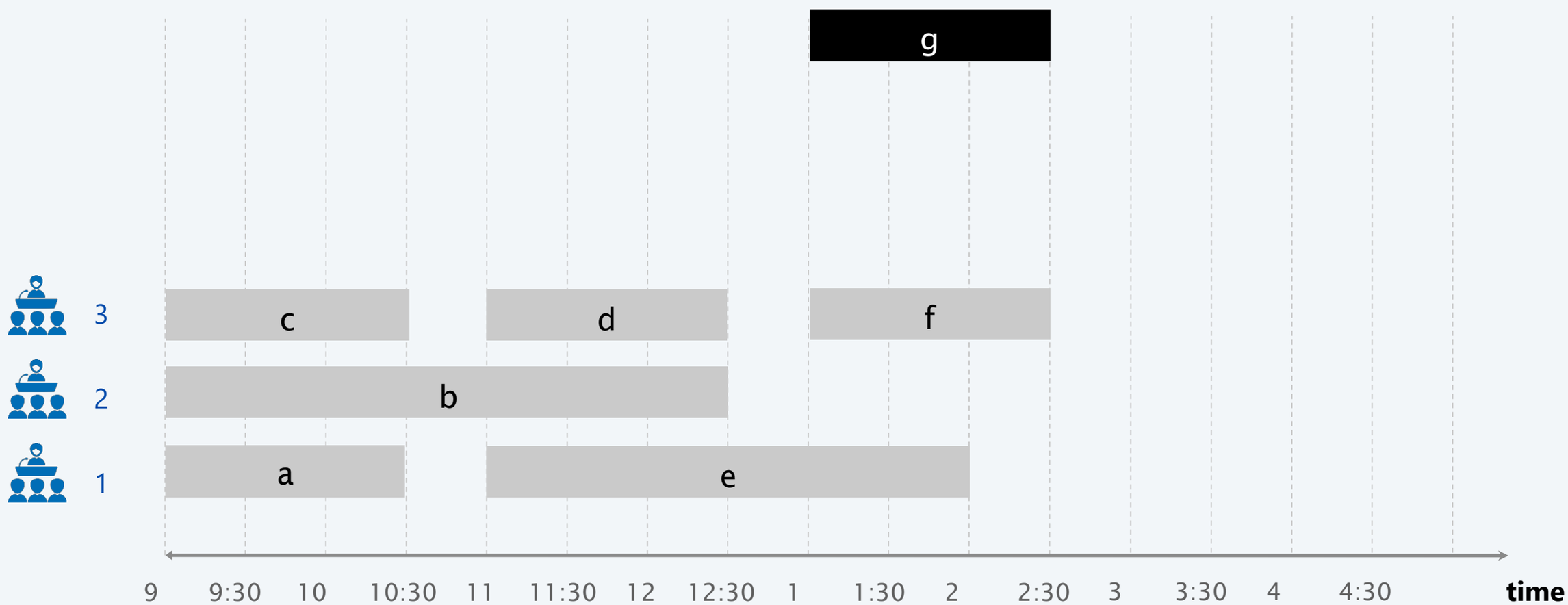


Earliest-start-time-first algorithm demo

Consider lectures in order of start time:

- Assign next lecture to any compatible classroom (if one exists).
- Otherwise, open up a new classroom.

lecture g is compatible with classroom 2



Earliest-start-time-first algorithm demo

Consider lectures in order of start time:

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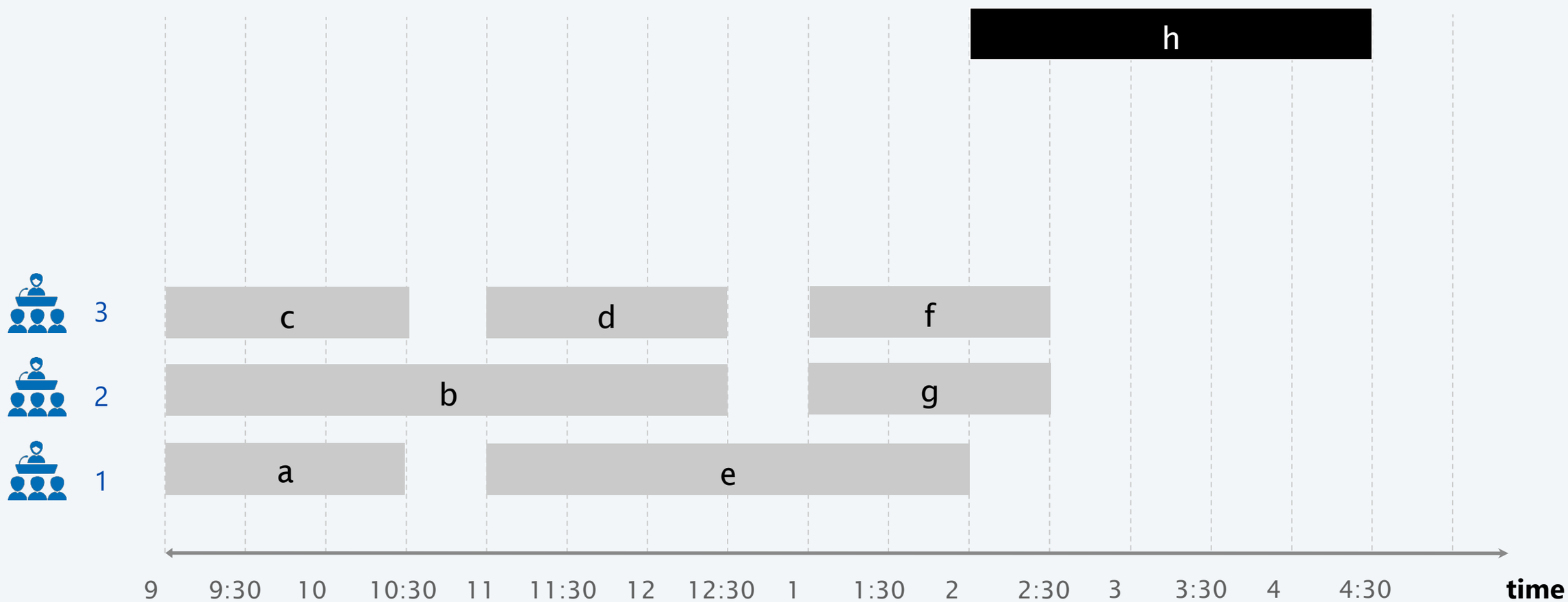
- Otherwise, open up a new classroom.

lecture h is compatible with classroom 1



Earliest-start-time-first algorithm demo

Consider lectures in order of start time:

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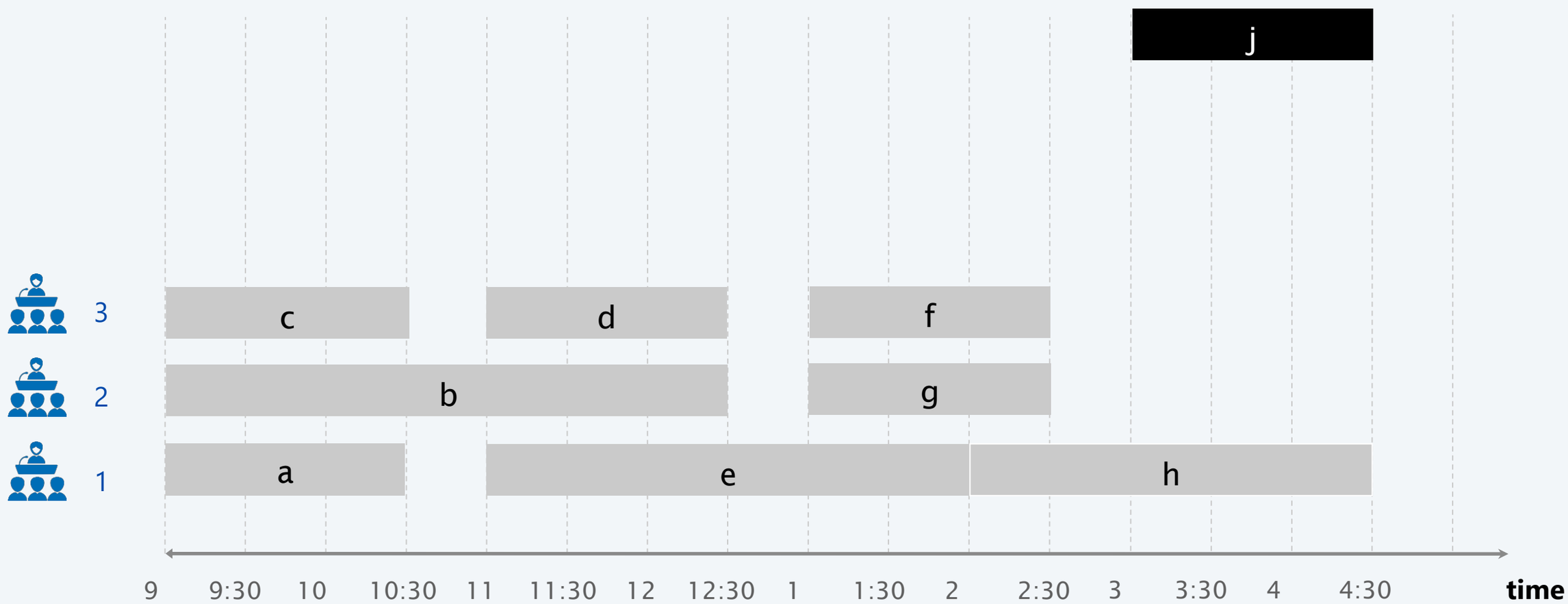
lecture j is compatible with classrooms 2 and 3



Earliest-start-time-first algorithm demo

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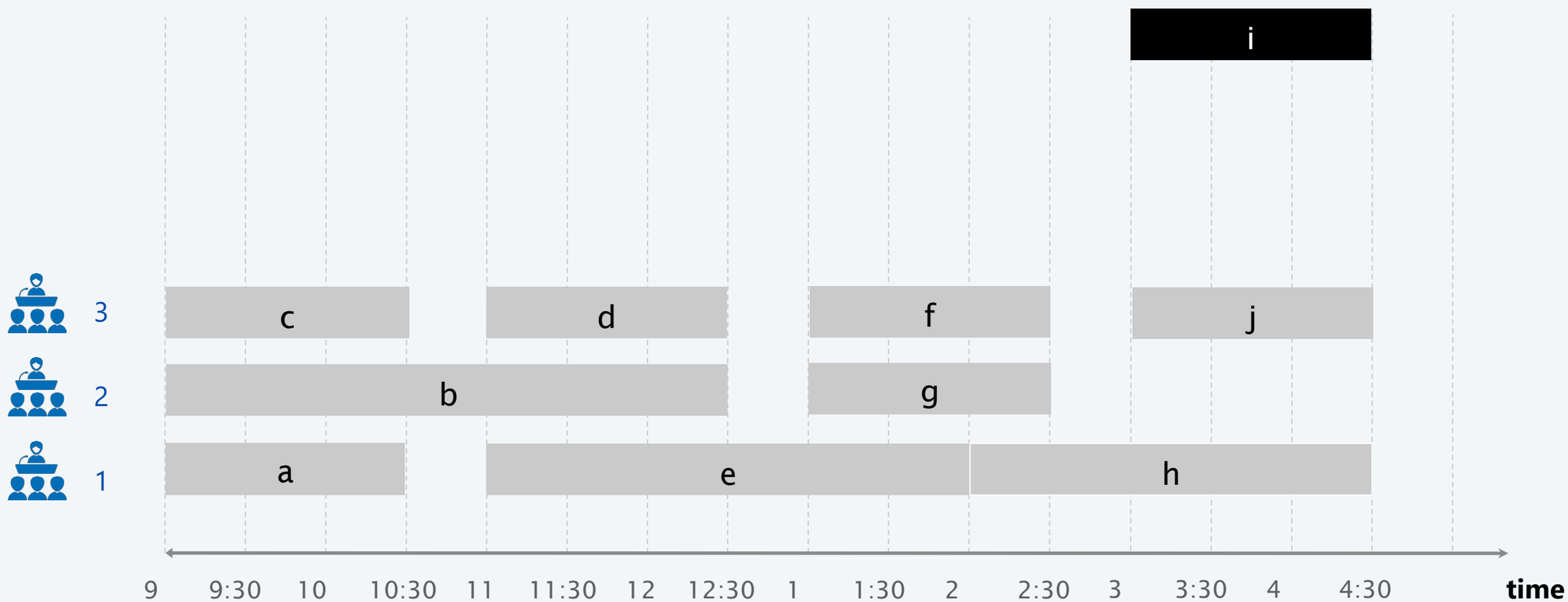

Earliest-start-time-first algorithm demo

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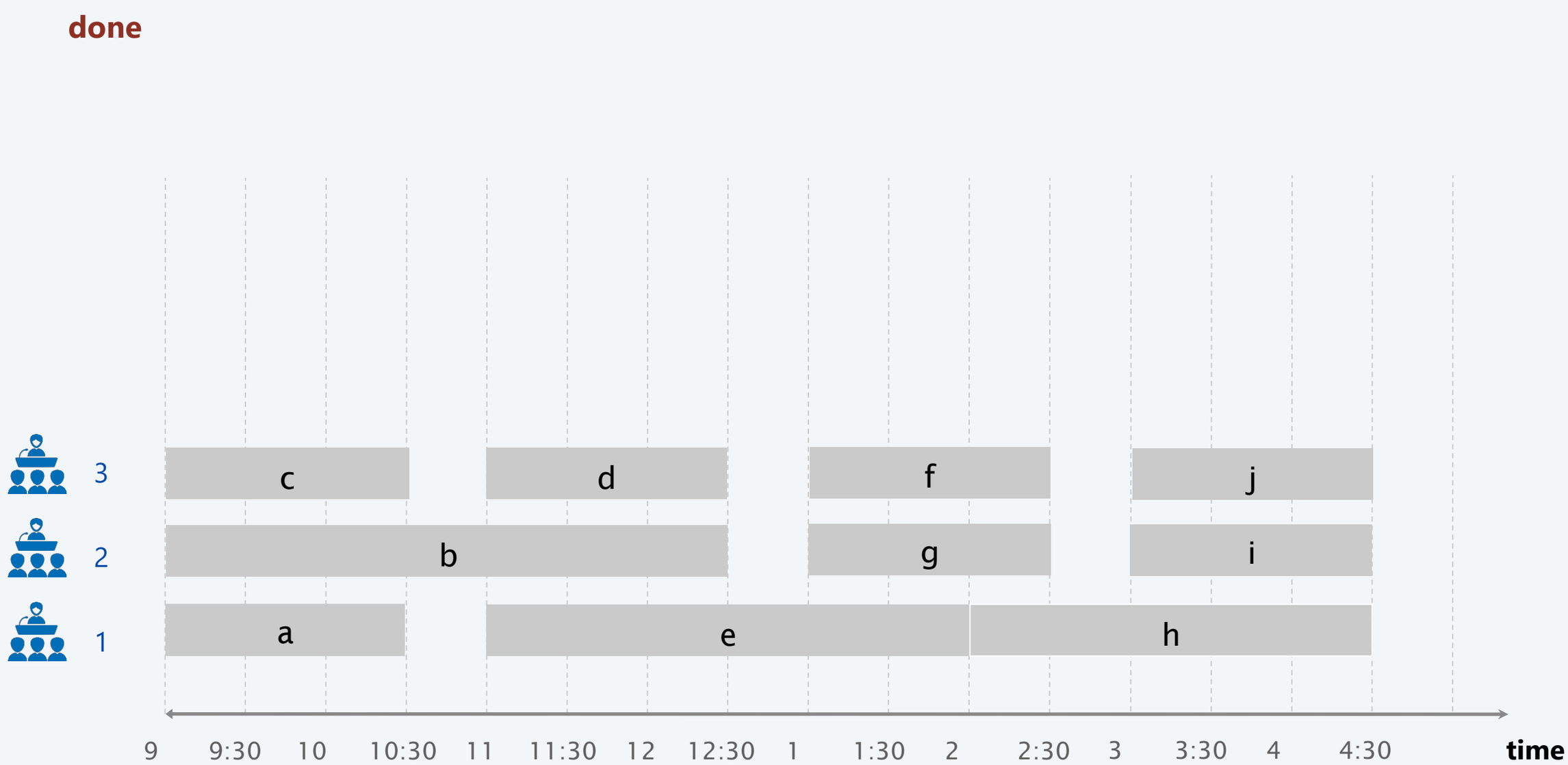


Interval partitioning: earliest-start-time-first algorithm

Proposition. The earliest-start-time-first algorithm can be implemented in $O(n \log n)$ time.

Pf.

- Sorting by start times takes $O(n \log n)$ time.
- Store classrooms in a priority queue (key = finish time of its last lecture).
  - to allocate a new classroom, INSERT classroom onto priority queue.
  - to schedule lecture j in classroom k, INCREASE-KEY of classroom k to f_j.
  - to determine whether lecture j is compatible with any classroom, compare s_j to FIND-MIN
- Total # of priority queue operations is $O(n)$; each takes $O(\log n)$ time. ▪

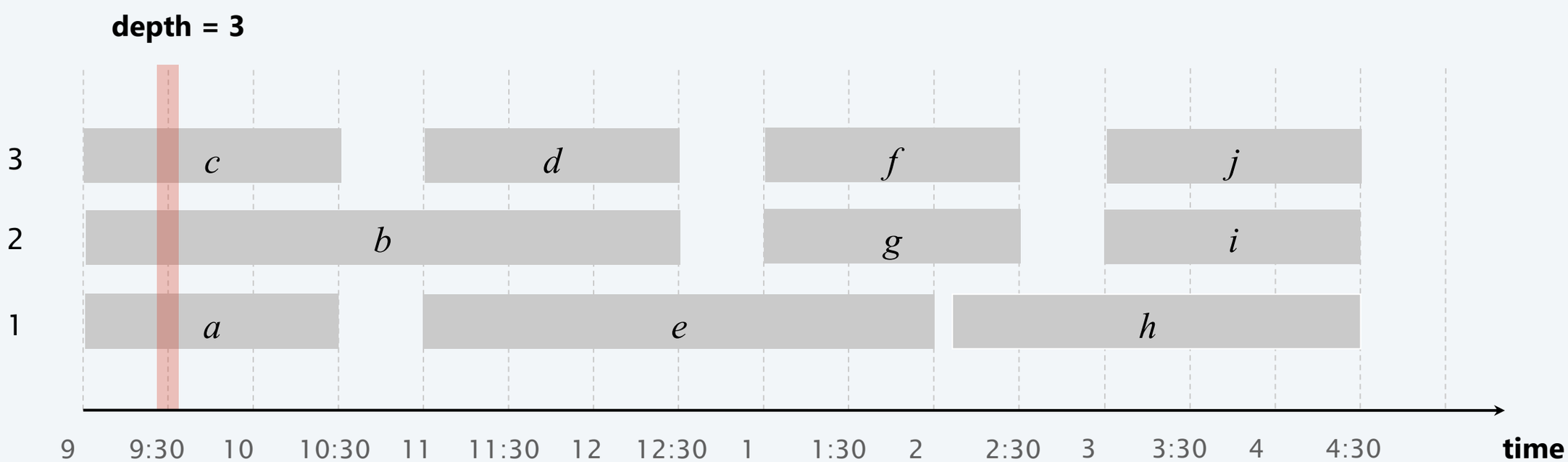
Remark. This implementation chooses a classroom k whose finish time of its last lecture is the earliest.

Interval partitioning: lower bound on optimal solution

Def. The depth of a set of open intervals is the maximum number of intervals that contain any given point.

Key observation. Number of classrooms needed $\geq$ depth.

Q. Does minimum number of classrooms needed always equal depth?
A. Yes! Moreover, earliest-start-time-first algorithm finds a schedule whose number of classrooms equals the depth.



Interval partitioning: analysis of earliest-start-time-first algorithm

Observation. The earliest-start-time first algorithm never schedules two incompatible lectures in the same classroom.

Theorem. Earliest-start-time-first algorithm is optimal.

Pf.

- Let d = number of classrooms that the algorithm allocates.
- Classroom d is opened because we needed to schedule a lecture, say j, that is incompatible with a lecture in each of $d-1$ other classrooms.
- Thus, these d lectures each end after s_j.
- Since we sorted by start time, each of these incompatible lectures start no later than s_j.
- Thus, we have d lectures overlapping at time $s_j + \varepsilon$.
- Key observation $\Rightarrow$ all schedules use $\geq d$ classrooms. ▪